



Convergence Analysis of Non-conforming Finite Element Method for a Quasi-static Contact Problem

Kamana Porwal¹ · Tanvi Wadhawan¹

Received: 16 April 2024 / Revised: 24 August 2024 / Accepted: 1 September 2024 /
Published online: 8 February 2025
© Shanghai University 2025

Abstract

We analyze the numerical solution of a non-linear evolutionary variational inequality, which is encountered in the investigation of quasi-static contact problems. The study of this article encompasses both the semi-discrete and fully discrete schemes, where we employ the backward Euler method for time discretization and utilize the lowest order Crouzeix-Raviart non-conforming finite-element method for spatial discretization. By assuming appropriate regularity conditions on the solution, we establish a priori error analysis for these schemes, achieving the optimal convergence order for linear elements. To illustrate the numerical convergence rates, we provide numerical results on a two-dimensional test problem.

Keywords Finite-element method · A priori error estimates · Variational inequalities · Quasi-static problem · Supremum norm

Mathematics Subject Classification 65N30 · 65N15

1 Introduction

Contact problems form the core of various mechanical structures and hold significant relevance in hydrostatics and thermostatics. These problems manifest in everyday scenarios, such as the interaction of braking pads with wheels, tires with roads, and pistons with skirts. Due to the significant importance of contact processes in various structural and mechanical systems, substantial efforts have been dedicated to their modeling and numerical simulations. The variational inequalities, often serving as variational formulations for these contact problems, have proven to be a potent tool in the mathematical analysis of these models (refer to, for instance, Refs. [6, 9, 20, 26, 27] and related references). In the article [20], an early attempt was made to investigate frictional contact problems using variational inequalities. For an extensive reference on the analysis and numerical approximation of contact problems concerning elastic materials, with and without friction, one can refer to Refs. [14, 17].

✉ Kamana Porwal
kamana@maths.iitd.ac.in

Tanvi Wadhawan
tanviwadhawan1234@gmail.com

¹ Department of Mathematics, Indian Institute of Technology, Delhi 110016, India

In the past decade, research has been extended to study contact problems involving viscoelastic and viscoplastic materials, giving rise to quasi-static contact problems which is a prototype of a time-dependent model. Quasi-static contact problems come into play when the applied forces on a system change gradually over time, resulting in negligible acceleration. We highlight the contributions that have enhanced the mathematical examination of quasi-static frictional contact, including findings from Refs. [2, 21, 22], which incorporate a normal compliance condition, and Ref. [11], which employs the regularization technique. The variational analysis of certain quasi-static contact problems within the context of linearized elasticity is available in references such as [1, 4, 21]. Quasi-static Signorini frictional problems utilizing a non-local Coulomb friction law have been examined in Refs. [10, 11]. On the other hand, in articles [3, 23, 24], the same problem is dealt with when a local Coulomb friction law is applied. The a priori and a posteriori error analyses of finite-element methods for linear viscoelastic problems without constraints are detailed in Ref. [15]. The numerical analysis of a bilateral frictional contact problem for the linear elastic material is carried out in Ref. [5]. In articles [12, 13], the discontinuous Galerkin (DG) formulation was devised for the gradient plasticity problem which is formulated as a quasi-static variational inequality of the second type. The article [28] focuses on a priori error analysis for quasi-static contact problems within a DG framework, considering both semi-discrete and fully discrete variational formulations. In their work, the authors attained optimal convergence rates when the solution is sufficiently regular and the consistency of the bilinear form is crucially used in carrying out the convergence analysis.

The main objective of this study is to conduct the numerical analysis of a quasi-static contact problem using the non-conforming finite-element method. In this context, we model the frictional process using the Tresca friction law, which includes a prescribed friction bound. This friction bound function represents the maximum magnitude of frictional traction before slip initiates. We investigate both spatially semi-discrete and fully discrete schemes for this problem, employing finite difference discretization in time and non-conforming finite-element discretization in space.

Mechanical Problem

The quasi-static contact model describes the evolution of the contact process between the linear elastic body and the rigid foundation. Let $\Omega \subset \mathbb{R}^2$ be the reference configuration of the linear elastic body with the boundary $\partial\Omega = \Gamma$. The boundary Γ is decomposed into three mutually disjoint, relatively open segments, the Dirichlet boundary Γ_D , the Neumann boundary Γ_N , and the potential contact boundary Γ_C with $\text{meas}(\Gamma_D) > 0$. The motion of the object is mathematically described using vector-valued functions $\mathbf{u}$ which depends both on position and time. Consequently, a material particle initially situated at the position $\mathbf{x} = (x_1, x_2)$ will find itself at the position $\mathbf{u}(\mathbf{x}, t)$ at time t . Let $\mathcal{I} = [0, T]$ be the time interval of consideration for some $T > 0$. The linearized strain tensor, denoted by $\epsilon(\mathbf{u}(\mathbf{x}, t))$ belongs to the class of second-order symmetric tensors $\mathbb{S}^2$ and is defined as

$$\epsilon(\mathbf{u}(\mathbf{x}, t)) := \frac{1}{2}(\nabla \mathbf{u}(\mathbf{x}, t) + (\nabla \mathbf{u}(\mathbf{x}, t))^T).$$

In this context, $t \in \mathcal{I}$ represents the time variable and $\mathbf{u}(\cdot, t): \Omega \rightarrow \mathbb{R}^2$ characterizes the displacement variable. The stress tensor $\sigma(\mathbf{u}(\mathbf{x}, t))$ obeys the following constitutive relation given by Hooke's law:

$$\sigma(\mathbf{u}(\mathbf{x}, t)) := \mathcal{C}\epsilon(\mathbf{u}(\mathbf{x}, t)), \quad (1)$$

where $\mathcal{C}: \mathbb{S}^2 \rightarrow \mathbb{S}^2$ is the bounded, symmetric, and positive definite fourth-order elasticity tensor. In the component form, the relation (1) can be rewritten as

$$\sigma_{ij} = \mathcal{C}_{ijkl} \epsilon_{kl}, \quad 1 \leq i, j, k, l \leq 2. \quad (2)$$

Given the homogeneity and isotropy of the linear elastic body, the fourth-order elasticity tensor can be explicitly expressed as follows:

$$\mathcal{C}_{ijkl} = \lambda \delta_{ij} \delta_{kl} + \mu (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}), \quad 1 \leq i, j, k, l \leq 2,$$

where λ and μ are the Lamé coefficients and δ_{ij} is the Kronecker delta. Thus, the stress tensor σ is given by

$$\sigma((u(x, t))) = \lambda(\text{tr}(\epsilon(u(x, t)))) \mathbf{Id} + 2\mu \epsilon(u(x, t)),$$

where $\mathbf{Id}$ denotes the identity matrix of order 2.

First, we shall recall the certain functional spaces that shall be relevant in the further analysis. To this end, let X denote a real Hilbert space with the norm $\|\cdot\|_X$. For $1 \leq p \leq \infty$, let $L^p(\mathcal{I}; X)$ denote the space of measurable functions $v: \mathcal{I} \rightarrow X$ endowed with the norm:

$$\|v\|_{L^p(\mathcal{I}; X)} = \begin{cases} \left(\int_{\mathcal{I}} \|v\|_X^p dt \right)^{\frac{1}{p}}, & \text{if } 1 \leq p < \infty, \\ \text{ess sup}_{t \in \mathcal{I}} \|v\|_X, & \text{if } p = \infty. \end{cases}$$

Further, for $k \in \mathbb{N}$, the Bochner space $W^{k,p}(\mathcal{I}; X)$ is defined as

$$W^{k,p}(\mathcal{I}; X) := \left\{ v \in L^p(\mathcal{I}; X) : \|v^{(j)}\|_{L^p(\mathcal{I}; X)} < \infty, \forall j \leq k \right\},$$

and is equipped with the norm:

$$\|v\|_{W^{k,p}(\mathcal{I}; X)} = \begin{cases} \left(\int_{\mathcal{I}} \sum_{j=1}^k \|v^{(j)}\|_X^p dt \right)^{\frac{1}{p}}, & \text{if } 1 \leq p < \infty, \\ \max_{0 \leq j \leq k} \text{ess sup}_{t \in \mathcal{I}} \|v^{(j)}\|_X, & \text{if } p = \infty. \end{cases}$$

Here, we use the notation $v^{(j)}$ to denote the distributional derivative of v of order $|j|$. Define $C(\mathcal{I}; X)$ to be the space of continuous functions $v: \mathcal{I} \rightarrow X$.

In this study, we distinguish vector-valued functions using bold symbols, while the scalar-valued functions are denoted in the conventional manner. To be precise, for any Hilbert space X , we define $\mathbf{X} := X \times X$. In addition, we suppress explicit dependence of the quantities on the spatial variable $\mathbf{x}$, i.e., we denote $\mathbf{u}(\mathbf{x}, t)$ by $\mathbf{u}(t)$ in the subsequent analysis. Let $\mathbf{n}$ represent the outward unit normal vector to Γ . Then, for any given vector $\mathbf{v}$, we can decompose it into its normal and tangential components as $v_n := \mathbf{v} \cdot \mathbf{n}$ and $\mathbf{v}_\tau := \mathbf{v} - v_n \mathbf{n}$, respectively. In a similar manner, for any tensor-valued function Ψ , we denote $\Psi_n := \Psi \mathbf{n} \cdot \mathbf{n}$ and $\Psi_\tau := \Psi \mathbf{n} - \Psi_n \mathbf{n}$ as its normal and tangential components, respectively. Furthermore, we will utilize the subsequent decomposition formula:

$$(\Psi \mathbf{n}) \cdot \mathbf{v} = \Psi_n v_n + \Psi_\tau \cdot \mathbf{v}_\tau. \quad (3)$$

The strong form associated with the quasi-static contact problem is to find the displacement vector $\mathbf{u}: \Omega \times [0, T] \rightarrow \mathbb{R}^2$ satisfying the following relations:

$$-\text{div } \sigma(\mathbf{u}(t)) = \mathbf{f}(t) \quad \text{in } \Omega \times (0, T), \quad (4)$$

$$\mathbf{u}(t) = \mathbf{0} \quad \text{on } \Gamma_D \times (0, T), \quad (5)$$

$$\boldsymbol{\sigma}(\mathbf{u}(t))\mathbf{n} = \mathbf{g}(t) \quad \text{on } \Gamma_N \times (0, T), \quad (6)$$

$$\mathbf{u}(\mathbf{x}, 0) = \mathbf{u}_o \quad \text{in } \Omega, \quad (7)$$

$$u_n = 0, \quad |\boldsymbol{\sigma}_\tau| \leq g_a \quad \text{on } \Gamma_C \times (0, T), \quad (8)$$

$$\left. \begin{aligned} |\boldsymbol{\sigma}_\tau| < g_a &\implies \dot{\mathbf{u}}_\tau = \mathbf{0} \\ |\boldsymbol{\sigma}_\tau| = g_a &\implies \dot{\mathbf{u}}_\tau = -\vartheta \boldsymbol{\sigma}_\tau \text{ for some } \vartheta \geq 0 \end{aligned} \right\} \quad \text{on } \Gamma_C \times (0, T), \quad (9)$$

where $\mathbf{f} \in \mathbf{W}^{1,\infty}(\mathcal{I}; \mathbf{L}^2(\Omega))$ and $\mathbf{g} \in \mathbf{W}^{1,\infty}(\mathcal{I}; \mathbf{L}^2(\Gamma_N))$ represent the density of the volume force and surface traction force, respectively. The stress tensor $\boldsymbol{\sigma}(\mathbf{u}(t))$ is defined in Eq. (1). Further, $g_a \in L^\infty(\Gamma_C)$ denotes the frictional bound function and $\dot{\mathbf{u}}$ represents the derivative with respect to time t .

Equation (4) refers to the equilibrium equation, signifying that the volume force of the density $\mathbf{f}(t)$ is acting in Ω . Equation (5) justifies that the displacement field vanishes on Γ_D indicating that the body is clamped within this region. A surface traction force with the density $\mathbf{g}(t)$ acts on region Γ_N , resulting in the relationship described in Eq. (6). The relation (7) describes the initial position of the body at time $t = 0$. The contact condition described in Eq. (9) is called the Tresca friction law. Therein, when the strict inequality is satisfied, the material point resides within the *stick zone*. In contrast, when the equality is met, the material point is in the *slip zone*. The boundary of these zones is not determined *a priori* and, therefore, is an inherent part of the problem.

Continuous Formulation of Quasi-static Contact Problem

In order to establish the variational formulation for the contact problem (4)–(9), we introduce the Hilbert space $\mathbf{V}$ as

$$\mathbf{V} := \left\{ \mathbf{v} \in \mathbf{H}^1(\Omega): \mathbf{v} = \mathbf{0} \text{ on } \Gamma_D, v_n = 0 \text{ on } \Gamma_C \right\}.$$

The variational formulation for the quasi-static contact problem is to find $\mathbf{u}: \mathcal{I} \rightarrow \mathbf{V}$ such that for a.e. $t \in \mathcal{I}$, the following holds:

$$a(\mathbf{u}(t), \mathbf{v} - \dot{\mathbf{u}}(t)) + j(\mathbf{v}) - j(\dot{\mathbf{u}}(t)) \geq (\mathbf{l}(t), \mathbf{v} - \dot{\mathbf{u}}(t)), \quad \forall \mathbf{v} \in \mathbf{V}, \quad (10)$$

$$\mathbf{u}(0) = \mathbf{u}_o, \quad (11)$$

where

$$\begin{aligned} a(\mathbf{v}, \mathbf{w}) &:= \int_{\Omega} \boldsymbol{\sigma}(\mathbf{v}) : \boldsymbol{\epsilon}(\mathbf{w}) \, dx, \\ j(\mathbf{v}) &:= \int_{\Gamma_C} g_a |\mathbf{v}_\tau| \, d\sigma, \\ (\mathbf{l}(t), \mathbf{v}) &:= \int_{\Omega} \mathbf{f}(t) \cdot \mathbf{v} \, dx + \int_{\Gamma_N} \mathbf{g}(t) \cdot \mathbf{v} \, d\sigma, \quad \forall \mathbf{v}, \mathbf{w} \in \mathbf{V}. \end{aligned}$$

Further, we assume the initial data satisfy $\mathbf{u}_o \in \mathbf{V}$ with

$$a(\mathbf{u}_o, \mathbf{v}) + j(\mathbf{v}) \geq (\mathbf{l}(0), \mathbf{v}), \quad \forall \mathbf{v} \in \mathbf{V}. \quad (12)$$

The existence and uniqueness of the solution $\mathbf{u}(t)$ for the problem (10)–(11) with the assumption (12) has been proved in the book [17]. In addition, the map $\psi: \mathbf{W}^{1,\infty}(\mathcal{I}; \mathbf{V}) \times \mathbf{V} \rightarrow \mathbf{L}^\infty(\mathcal{I}; \mathbf{V})$ defined by

$$\psi(\mathbf{l}(t), \mathbf{u}_o) \longrightarrow \mathbf{u}$$

is a Lipschitz continuous function from $W^{1,\infty}(\mathcal{I}; V) \times V \longrightarrow L^\infty(\mathcal{I}; V)$.

The rest of the article is structured as follows: Sect. 2 begins by introducing the fundamental notations. Following that, we define the discrete space on which the discrete problem is formulated. Therein, we also introduce the Crouzeix-Raviart interpolation operator, denoted by $\mathbf{I}^{CR}$. This operator exhibits commutativity with the time derivative which is a pivotal aspect of the error analysis. The subsequent sections, namely Sects. 3 and 4, are devoted to establishing a priori error estimates for the semi-discrete scheme and the fully discrete scheme, respectively. Finally, in Sect. 5, we present the results of a numerical experiment designed to illustrate the theoretical findings and results discussed in the preceding sections.

2 Discrete Problem

In this segment, we first list the notations which will be useful in the forthcoming analysis.

- $\mathcal{T}_h$ denotes a regular simplicial triangulation of the domain Ω ,
- T refers to an element of $\mathcal{T}_h$,
- $\mathcal{V}_T$ refers to the set of all vertices of the triangle T ,
- $\mathcal{V}_h$ refers to the set of vertices of $\mathcal{T}_h$,
- $\mathcal{V}_e$ refers to the set of vertices lying on the edge e ,
- $\mathcal{E}_h$ is the set of edges of $\mathcal{T}_h$,
- $\mathcal{E}_h^i$ refers to the set of edges lying in the interior of $\mathcal{T}_h$,
- $\mathcal{E}_h^b$ refers to the set of edges lying on the boundary of $\mathcal{T}_h$,
- $\mathcal{E}_h^D$ refers to the set of edges lying on Γ_D ,
- $\mathcal{E}_h^C$ refers to the set of edges lying on Γ_C ,
- $\mathcal{E}_h^0 := \mathcal{E}_h^i \cup \mathcal{E}_h^D$,
- $\mathcal{V}_h^D$ refers to the vertices lying on $\overline{\Gamma}_D$,
- $\mathcal{V}_h^N$ refers to the vertices lying on Γ_N ,
- $\mathcal{V}_h^C$ refers to the vertices lying on $\overline{\Gamma}_C$,
- $\mathcal{M}_h^D$ refers to the set of midpoints of edges lying on Γ_D ,
- $\mathcal{M}_h^C$ refers to the set of midpoints of edges lying on Γ_C ,
- $\mathcal{E}_p$ denotes the set of edges sharing the node p ,
- $\mathcal{T}_p$ represents the collection of all triangles sharing the node p ,
- $\mathcal{T}_e$ denotes the set of triangles sharing the edge e ,
- h_T is the diameter of the triangle T ,
- $h := \max \{h_T : T \in \mathcal{T}_h\}$,
- h_e is the length of an edge e ,
- $\mathbb{P}_k(T)$ represents the space of polynomials of the global degree $\leq k$ on each triangle T where $0 \leq k \in \mathbb{Z}$,
- $|S|$ represents the number of elements in the set S .

Throughout the analysis, we assume C as a positive constant independent of the mesh parameter h . In addition, we use the notation $X \lesssim Y$ to represent $X \leq CY$.

In order to deal with discontinuous functions conveniently, we introduce the broken Sobolev space $H^1(\Omega, \mathcal{T}_h)$ as follows:

$$H^1(\Omega, \mathcal{T}_h) := \left\{ v \in L^2(\Omega) : v|_T \in H^1(T), \forall T \in \mathcal{T}_h \right\}.$$

In the sequel, we will define jumps and averages of scalar, vector, and tensor-valued functions. To this end, let $e \in \mathcal{E}_h^i$ be an arbitrary interior edge shared by two adjacent elements T_1 and T_2 , i.e., $e \in \partial T_1 \cap \partial T_2$. Denote by $\mathbf{n}_{T_1}$, the outward unit normal vector of e pointing from T_1 to T_2 and set $\mathbf{n}_{T_2} = -\mathbf{n}_{T_1}$. The jumps $[\![\cdot]\!]$ and averages $\{\!\{ \cdot \}\!\}$ are defined in the following way:

- for the function $w \in H^1(\Omega, \mathcal{T}_h)$, define

$$\{\!\{ w \}\!\} := \frac{w|_{T_1} + w|_{T_2}}{2}, \quad [\![w]\!] := w|_{T_1} \mathbf{n}_{T_1} + w|_{T_2} \mathbf{n}_{T_2};$$

- for any function $\mathbf{v} \in [H^1(\Omega, \mathcal{T}_h)]^2$, define

$$\{\!\{ \mathbf{v} \}\!\} := \frac{\mathbf{v}|_{T_1} + \mathbf{v}|_{T_2}}{2}, \quad [\![\mathbf{v}]\!] := \mathbf{v}|_{T_1} \otimes \mathbf{n}_{T_1} + \mathbf{v}|_{T_2} \otimes \mathbf{n}_{T_2};$$

- for the function $\phi \in [H^1(\Omega, \mathcal{T}_h)]^{2 \times 2}$, set

$$\{\!\{ \phi \}\!\} := \frac{\phi|_{T_1} + \phi|_{T_2}}{2}, \quad [\![\phi]\!] := \phi|_{T_1} \mathbf{n}_{T_1} + \phi|_{T_2} \mathbf{n}_{T_2}.$$

Here, the notation $\mathbf{a} \otimes \mathbf{b}$ is used to describe the dyadic product of two vectors $\mathbf{a}, \mathbf{b} \in \mathbb{R}^2$, i.e., $(\mathbf{a} \otimes \mathbf{b})_{ij} := a_i b_j$. Analogously, for the notational convenience, we define jumps and averages for the scalar-valued function $w \in H^1(\Omega, \mathcal{T}_h)$, the vector-valued function $\mathbf{v} \in \mathbf{H}^1(\Omega, \mathcal{T}_h)$ and the tensor-valued function $\phi \in [H^1(\Omega, \mathcal{T}_h)]^{2 \times 2}$ on the boundary edges $e \in \mathcal{E}_h^b$ as follows:

$$\begin{aligned} \{\!\{ w \}\!\} &:= w, & [\![w]\!] &:= w \mathbf{n}_e; \\ \{\!\{ \mathbf{v} \}\!\} &:= \mathbf{v}, & [\![\mathbf{v}]\!] &:= \mathbf{v} \otimes \mathbf{n}_e; \\ \{\!\{ \phi \}\!\} &:= \phi, & [\![\phi]\!] &:= \phi \mathbf{n}_e; \end{aligned}$$

where $T \in \mathcal{T}_h$ contains the edge e and $\mathbf{n}_e$ is the outward unit normal on the edge e pointing outside T .

Next, we introduce the Crouzeix-Raviart finite-element space to discretize problem (10)–(11) as

$$\begin{aligned} V_{CR}^h := & \left\{ \mathbf{v}_h \in L^2(\Omega) : \mathbf{v}_h|_T \in \mathbb{P}_1(T), \quad \forall T \in \mathcal{T}_h, \int_e [\![\mathbf{v}_h]\!] \, d\sigma = 0, \quad \forall e \in \mathcal{E}_h^i \right. \\ & \left. \text{and } \mathbf{v}_h(m_e) = 0, \quad \forall m_e \in \mathcal{M}_h^D \right\} \end{aligned} \quad (13)$$

and

$$\mathcal{K}_{CR}^h := \left\{ \mathbf{v}_h \in V_{CR}^h(\mathcal{T}_h) : v_{hn}(m_e) = 0, \quad \forall m_e \in \mathcal{M}_h^C \right\}, \quad (14)$$

where $\mathbf{n}$ represents the outward unit normal vector on Γ_C .

At the several occasions, we will be using the following inverse and trace inequalities.

- **Trace inequality** [8, 25]

Lemma 1 Let $T \in \mathcal{T}_h$ and e be an edge of T . Then for $\phi \in W^{1,p}(T)$, $1 \leq p < \infty$, the following holds:

$$\|\phi\|_{L^p(e)}^p \lesssim h_e^{-1} \left(\|\phi\|_{L^p(T)}^p + h_e^p \|\nabla \phi\|_{L^p(T)}^p \right).$$

- **Inverse inequalities** [8, 25]

Lemma 2 Let $1 \leq s, t \leq \infty$ and $\mathbf{v}_h \in V_{CR}^h$. Then, it holds that

- (i) $\|\mathbf{v}_h\|_{W^{m,s}(T)} \lesssim h_T^{l-m} h_T^{2(\frac{1}{s}-\frac{1}{l})} \|\mathbf{v}_h\|_{W^{l,l}(T)}, \quad \forall T \in \mathcal{T}_h, \quad l \leq m,$
- (ii) $\|\mathbf{v}_h\|_{L^\infty(T)} \lesssim h_T^{-1} \|\mathbf{v}_h\|_{L^2(T)}, \quad \forall T \in \mathcal{T}_h,$
- (iii) $\|\mathbf{v}_h\|_{L^\infty(e)} \lesssim h_e^{-\frac{1}{2}} \|\mathbf{v}_h\|_{L^2(e)}, \quad \forall e \in \mathcal{E}_h.$

Next, we will define the following interpolation operator which will play a crucial role in the upcoming analysis. Define the interpolation operator $\mathbf{I}^{CR}: \mathbf{H}^1(\Omega) \rightarrow \mathbf{V}_{CR}^h$ by

$$\mathbf{I}^{CR}(\mathbf{v}) := \sum_{e \in \mathcal{E}_h} \left(\int_e \frac{1}{h_e} v_1 \, d\sigma \, \psi_e, \int_e \frac{1}{h_e} v_2 \, d\sigma \, \psi_e \right),$$

where ψ_e refers to the edge oriented basis function for the space $\mathbf{V}_{CR}^h$ which satisfies the following properties:

- (i) ψ_e should be equal to one along edge e ;
- (ii) $\psi_e(m_{e'}) = 0$ for any other edge $e' \in \mathcal{E}^h \setminus e$;
- (iii) support of ψ_e is contained in $\mathcal{T}_e$.

The key characteristic of this map is that it commutes with the time derivative, i.e.,

$$\begin{aligned} \frac{\partial}{\partial t} \mathbf{I}^{CR}(\mathbf{v}) &= \frac{\partial}{\partial t} \sum_{e \in \mathcal{E}_h} \left(\int_e \frac{1}{h_e} v_1 \, d\sigma \, \psi_e, \int_e \frac{1}{h_e} v_2 \, d\sigma \, \psi_e \right) \\ &= \sum_{e \in \mathcal{E}_h} \left(\int_e \frac{1}{h_e} \frac{\partial v_1}{\partial t} \, d\sigma \, \psi_e, \int_e \frac{1}{h_e} \frac{\partial v_2}{\partial t} \, d\sigma \, \psi_e \right) \\ &= \mathbf{I}^{CR} \left(\frac{\partial \mathbf{v}}{\partial t} \right). \end{aligned}$$

Next, we will see the approximation properties of the interpolation map $\mathbf{I}^{CR}$ in the next lemma [7, 8].

Lemma 3 For any $\mathbf{v} \in \mathbf{H}^v(T)$, $T \in \mathcal{T}_h$, the following holds:

$$|\mathbf{v} - \mathbf{I}^{CR}(\mathbf{v})|_{\mathbf{H}^l(T)} \lesssim h_T^{v-l} |\mathbf{v}|_{\mathbf{H}^v(T)} \quad \text{for } 0 \leq l \leq v,$$

where $1 \leq v \leq 2$.

Note that for the sake of notational convenience, we denote $\mathbf{I}^{CR}(\mathbf{u}(t))$ by $\mathbf{u}_I(t)$ in the upcoming sections.

In this analysis, we introduce and examine two types of approximation schemes. Discretizing only the spatial variables yields semi-discrete schemes, whereas discretizing the spatial and temporal variables results in fully discrete schemes. For both schemes, we aim to develop rigorous *a priori* error analysis and achieve optimal order of convergence under the additional assumptions on the solution's regularity.

3 Semi-discrete Approximation

The spatially semi-discrete formulation for the variational inequality (10)–(11) is given by

Problem 1 (Semi-Discrete Variational Inequality) Find $\mathbf{u}_h: \mathcal{I} \rightarrow \mathcal{K}_{CR}^h$ satisfying

$$a_h(\mathbf{u}_h(t), \mathbf{v}_h - \dot{\mathbf{u}}_h(t)) + j(\mathbf{v}_h) - j(\dot{\mathbf{u}}_h(t)) \geq (\mathbf{I}(t), \mathbf{v}_h - \dot{\mathbf{u}}_h(t)), \quad \forall \mathbf{v}_h \in \mathcal{K}_{CR}^h, \quad (15)$$

$$\mathbf{u}_h(0) = \mathbf{I}^{CR} \mathbf{u}_o. \quad (16)$$

Here, the discrete bilinear form $a_h: \mathbf{V}_{CR}^h \times \mathbf{V}_{CR}^h \rightarrow \mathbb{R}$ is defined as

$$\begin{aligned} a_h(\mathbf{v}_h, \mathbf{w}_h) &:= \sum_{T \in \mathcal{T}_h} \int_T \boldsymbol{\sigma}(\mathbf{v}_h) : \boldsymbol{\epsilon}(\mathbf{w}_h) \, dx \\ &\quad + \sum_{e \in \mathcal{E}_h^0} 2\rho\mu \int_e \frac{1}{h_e} \llbracket \mathbf{v}_h \rrbracket : \llbracket \mathbf{w}_h \rrbracket \, d\sigma. \end{aligned}$$

Note that the parameter $\rho > 0$ is introduced in the discrete bilinear form $a_h(\cdot, \cdot)$ in order to stabilize it [18]. Next, we define the norm $\|\cdot\|$ on the space $\mathbf{V}_{CR}^h + \mathbf{V} \cap [H^2(\Omega)]^2$ using the following relations:

$$\|\mathbf{v}\|^2 := |\mathbf{v}|_h^2 + |\mathbf{v}|_*^2,$$

where

$$|\mathbf{v}|_h^2 := \sum_{T \in \mathcal{T}_h} |\mathbf{v}|_T^2, \quad |\mathbf{v}|_*^2 := \sum_{e \in \mathcal{E}_h^0} \frac{2\rho\mu}{h_e} \|\llbracket \mathbf{v} \rrbracket\|_{L^2(e)}^2$$

with

$$|\mathbf{v}|_T^2 := \int_T \boldsymbol{\sigma}(\mathbf{v}) : \boldsymbol{\epsilon}(\mathbf{v}) \, dx, \quad \|\llbracket \mathbf{v} \rrbracket\|_{L^2(e)}^2 := \int_e \llbracket \mathbf{v} \rrbracket : \llbracket \mathbf{v} \rrbracket \, d\sigma.$$

Note that the bilinear form $a_h(\cdot, \cdot)$ is continuous and coercive with respect to the norm $\|\cdot\|$. Thus, the unique solvability of Problem 1 follows from the Stampacchia theorem [17].

3.1 Convergence Analysis for Spatially Semi-discrete Approximation

In the next segment, we aim to derive *a priori* error estimates for the semi-discrete variational inequality (15)–(16) taking into account appropriate regularity assumptions on the exact solution $\mathbf{u}(t)$. The frictional contact condition results in a singular behavior of $\mathbf{u}(t)$ near the free boundary around Γ_C , even when the forces $\mathbf{f}(t)$ and $\mathbf{g}(t)$ exhibit sufficient regularity. Keeping this in mind, it is realistic to assume that $\mathbf{u}(t)$ and $\dot{\mathbf{u}}(t)$ belong to $L^\infty[\mathcal{I}; \mathbf{H}^2(\Omega)]$. The following theorem guarantees optimal order *a priori* error estimates for the quasi-static contact problem. Therein, the error is measured in the $L^\infty(\mathcal{I}; \mathbf{V}_h)$ norm and it is shown to decay with order $\mathcal{O}(h)$.

Theorem 1 *Let $\mathbf{u}(t)$ be the solution of the continuous problem (10)–(11) and $\mathbf{u}_h(t)$ be the solution of the semi-discrete problem (15)–(16). Assume $\mathbf{u}, \dot{\mathbf{u}} \in L^\infty[\mathcal{I}; \mathbf{H}^2(\Omega)]$. Then, we have*

$$\|\mathbf{u} - \mathbf{u}_h\|_{L^\infty(\mathcal{I}; \mathbf{V}_h)} \lesssim h.$$

Proof Let $\mathbf{e}(t) = \mathbf{u}(t) - \mathbf{u}_h(t) = \boldsymbol{\theta}(t) + \boldsymbol{\eta}(t)$, where $\boldsymbol{\theta}(t) := \mathbf{u}(t) - \mathbf{u}_I(t)$ and $\boldsymbol{\eta}(t) := \mathbf{u}_I(t) - \mathbf{u}_h(t)$. Now, employing the triangle inequality, we obtain

$$\|\mathbf{e}(t)\| \leq \|\boldsymbol{\theta}(t)\| + \|\boldsymbol{\eta}(t)\|.$$

First, we will derive an upper bound for $\|\boldsymbol{\theta}(t)\|$:

$$\|\boldsymbol{\theta}(t)\|^2 \lesssim \sum_{T \in \mathcal{T}_h} |\mathbf{u}(t) - \mathbf{u}_I(t)|_{\mathbf{H}^1(T)}^2 + \sum_{e \in \mathcal{E}_h^0} \frac{2\rho\mu}{h_e} \|\llbracket \mathbf{u}(t) - \mathbf{u}_I(t) \rrbracket\|_{L^2(e)}^2.$$

Applying the trace inequality and properties of the interpolation operator (as detailed in Lemma 3), we get

$$\|\theta(t)\|^2 \lesssim \sum_{T \in \mathcal{T}_h} h_T^2 |\mathbf{u}(t)|_{\mathbf{H}^2(T)}^2.$$

Next, we need to bound $\|\mathbf{u}_I(t) - \mathbf{u}_h(t)\|$. To this end, we have

$$\begin{aligned} \frac{1}{2} \frac{\partial}{\partial t} \|\mathbf{u}_I(t) - \mathbf{u}_h(t)\|^2 &= \frac{1}{2} \frac{\partial}{\partial t} \left(\sum_{T \in \mathcal{T}_h} \int_T \boldsymbol{\sigma}(\mathbf{u}_I(t) - \mathbf{u}_h(t)) : \boldsymbol{\epsilon}(\mathbf{u}_I(t) - \mathbf{u}_h(t)) \, dx \right. \\ &\quad \left. + \sum_{e \in \mathcal{E}_h^0} \frac{2\rho\mu}{h_e} \int_e [\![\mathbf{u}_I(t) - \mathbf{u}_h(t)]\!] : [\![\mathbf{u}_I(t) - \mathbf{u}_h(t)]\!] \, d\sigma \right) \\ &= a_h(\mathbf{u}_I(t) - \mathbf{u}_h(t), \dot{\mathbf{u}}_I(t) - \dot{\mathbf{u}}_h(t)) \\ &= a_h(\mathbf{u}_I(t) - \mathbf{u}(t), \dot{\mathbf{u}}_I(t) - \dot{\mathbf{u}}_h(t)) + a_h(\mathbf{u}(t), \dot{\mathbf{u}}_I(t) - \dot{\mathbf{u}}_h(t)) \\ &\quad - a_h(\mathbf{u}_h(t), \dot{\mathbf{u}}_I(t) - \dot{\mathbf{u}}_h(t)). \end{aligned} \quad (17)$$

Using the fact that $\dot{\mathbf{u}}_I(t) \in \mathcal{K}_{CR}^h$, we find

$$a_h(\mathbf{u}_h(t), \dot{\mathbf{u}}_I(t) - \dot{\mathbf{u}}_h(t)) + j(\dot{\mathbf{u}}_I(t)) - j(\dot{\mathbf{u}}_h(t)) \geq (\mathbf{l}(t), \dot{\mathbf{u}}_I(t) - \dot{\mathbf{u}}_h(t)).$$

Thus,

$$-a_h(\mathbf{u}_h(t), \dot{\mathbf{u}}_I(t) - \dot{\mathbf{u}}_h(t)) \leq j(\dot{\mathbf{u}}_I(t)) - j(\dot{\mathbf{u}}_h(t)) - (\mathbf{l}(t), \dot{\mathbf{u}}_I(t) - \dot{\mathbf{u}}_h(t)). \quad (18)$$

Inserting Eq. (18) in the relation (17), we obtain

$$\begin{aligned} \frac{1}{2} \frac{\partial}{\partial t} \|\mathbf{u}_I(t) - \mathbf{u}_h(t)\|^2 &\leq a_h(\mathbf{u}_I(t) - \mathbf{u}(t), \dot{\mathbf{u}}_I(t) - \dot{\mathbf{u}}_h(t)) \\ &\quad + a_h(\mathbf{u}(t), \dot{\mathbf{u}}_I(t) - \dot{\mathbf{u}}_h(t)) + j(\dot{\mathbf{u}}_I(t)) \\ &\quad - j(\dot{\mathbf{u}}_h(t)) - (\mathbf{l}(t), \dot{\mathbf{u}}_I(t) - \dot{\mathbf{u}}_h(t)). \end{aligned} \quad (19)$$

Applying integration by parts to the second term of the last relation, we obtain

$$\begin{aligned} a_h(\mathbf{u}(t), \dot{\mathbf{u}}_I(t) - \dot{\mathbf{u}}_h(t)) &= \sum_{T \in \mathcal{T}_h} \int_T \boldsymbol{\sigma}(\mathbf{u}(t)) : \boldsymbol{\epsilon}(\dot{\mathbf{u}}_I(t) - \dot{\mathbf{u}}_h(t)) \, dx \\ &= \sum_{T \in \mathcal{T}_h} \int_T \mathbf{f}(t) \cdot (\dot{\mathbf{u}}_I(t) - \dot{\mathbf{u}}_h(t)) \, dx \\ &\quad + \sum_{e \in \mathcal{E}_h^N} \int_e \mathbf{g}(t) \cdot (\dot{\mathbf{u}}_I(t) - \dot{\mathbf{u}}_h(t)) \, d\sigma \\ &\quad + \sum_{e \in \mathcal{E}_h^C} \int_e (\boldsymbol{\sigma}(\mathbf{u}(t))\mathbf{n}) \cdot (\dot{\mathbf{u}}_I(t) - \dot{\mathbf{u}}_h(t)) \, d\sigma \\ &\quad + \sum_{e \in \mathcal{E}_h^0} \int_e [\![\boldsymbol{\sigma}(\mathbf{u}(t))]\!] : [\![\dot{\mathbf{u}}_I(t) - \dot{\mathbf{u}}_h(t)]\!] \, d\sigma. \end{aligned} \quad (20)$$

Clubbing Eq. (20) in Eq. (19), we find

$$\frac{1}{2} \frac{\partial}{\partial t} \|\mathbf{u}_I(t) - \mathbf{u}_h(t)\|^2 \leq a_h(\mathbf{u}_I(t) - \mathbf{u}(t), \dot{\mathbf{u}}_I(t) - \dot{\mathbf{u}}_h(t))$$

$$\begin{aligned}
& + j(\dot{\mathbf{u}}_I(t)) - j(\dot{\mathbf{u}}_h(t)) + \sum_{e \in \mathcal{E}_h^C} \int_e (\boldsymbol{\sigma}(\mathbf{u}(t))\mathbf{n}) \cdot (\dot{\mathbf{u}}_I(t) - \dot{\mathbf{u}}_h(t)) \, d\sigma \\
& + \sum_{e \in \mathcal{E}_h^0} \int_e \{\{\boldsymbol{\sigma}(\mathbf{u}(t))\}\} : [\![\dot{\mathbf{u}}_I(t) - \dot{\mathbf{u}}_h(t)]\!] \, d\sigma.
\end{aligned}$$

Integrating the last inequality from 0 to t , we obtain

$$\begin{aligned}
\|\mathbf{u}_I(t) - \mathbf{u}_h(t)\|^2 & \leq \|\mathbf{u}_I(0) - \mathbf{u}_h(0)\|^2 + 2 \int_0^t a_h(\mathbf{u}_I(s) - \mathbf{u}(s), \dot{\mathbf{u}}_I(s) - \dot{\mathbf{u}}_h(s)) \, ds \\
& + 2 \int_0^t (j(\dot{\mathbf{u}}_I(s)) - j(\dot{\mathbf{u}}_h(s))) \, ds \\
& + 2 \int_0^t \sum_{e \in \mathcal{E}_h^C} \int_e (\boldsymbol{\sigma}(\mathbf{u}(s))\mathbf{n}) \cdot (\dot{\mathbf{u}}_I(s) - \dot{\mathbf{u}}_h(s)) \, d\sigma \, ds \\
& + 2 \int_0^t \sum_{e \in \mathcal{E}_h^0} \int_e \{\{\boldsymbol{\sigma}(\mathbf{u}(s))\}\} : [\![\dot{\mathbf{u}}_I(s) - \dot{\mathbf{u}}_h(s)]\!] \, d\sigma \, ds.
\end{aligned}$$

Exploiting the fact that $\mathbf{u}_I(0) = \mathbf{u}_h(0)$, we have

$$\begin{aligned}
\|\mathbf{u}_I(t) - \mathbf{u}_h(t)\|^2 & \leq 2 \int_0^t a_h(\mathbf{u}_I(s) - \mathbf{u}(s), \dot{\mathbf{u}}_I(s) - \dot{\mathbf{u}}_h(s)) \, ds \\
& + 2 \int_0^t \sum_{e \in \mathcal{E}_h^C} \int_e \sigma_n(\mathbf{u}(s)) \cdot (\dot{\mathbf{u}}_I(s) - \dot{\mathbf{u}}_h(s))_n \, d\sigma \, ds \\
& + 2 \int_0^t \sum_{e \in \mathcal{E}_h^C} \int_e \sigma_\tau(\mathbf{u}(s)) \cdot (\dot{\mathbf{u}}_I(s) - \dot{\mathbf{u}}_h(s))_\tau \, d\sigma \, ds \\
& + 2 \int_0^t (j(\dot{\mathbf{u}}_I(s)) - j(\dot{\mathbf{u}}_h(s))) \, ds \\
& + 2 \int_0^t \sum_{e \in \mathcal{E}_h^0} \int_e \{\{\boldsymbol{\sigma}(\mathbf{u}(s))\}\} : [\![\dot{\mathbf{u}}_I(s) - \dot{\mathbf{u}}_h(s)]\!] \, d\sigma \, ds \\
& = A_1 + A_2 + A_3 + A_4,
\end{aligned} \tag{21}$$

where

$$\begin{aligned}
A_1 & = 2 \int_0^t a_h(\mathbf{u}_I(s) - \mathbf{u}(s), \dot{\mathbf{u}}_I(s) - \dot{\mathbf{u}}_h(s)) \, ds, \\
A_2 & = 2 \int_0^t \sum_{e \in \mathcal{E}_h^C} \int_e \sigma_\tau(\mathbf{u}(s)) \cdot (\dot{\mathbf{u}}_I(s) - \dot{\mathbf{u}}_h(s))_\tau \, d\sigma \, ds + 2 \int_0^t (j(\dot{\mathbf{u}}_I(s)) - j(\dot{\mathbf{u}}_h(s))) \, ds, \\
A_3 & = 2 \int_0^t \sum_{e \in \mathcal{E}_h^0} \int_e \{\{\boldsymbol{\sigma}(\mathbf{u}(s))\}\} : [\![\dot{\mathbf{u}}_I(s) - \dot{\mathbf{u}}_h(s)]\!] \, d\sigma \, ds, \\
A_4 & = 2 \int_0^t \sum_{e \in \mathcal{E}_h^C} \int_e \sigma_n(\mathbf{u}(s)) \cdot (\dot{\mathbf{u}}_I(s) - \dot{\mathbf{u}}_h(s))_n \, d\sigma \, ds.
\end{aligned}$$

We will now estimate each term individually. To establish an upper bound for A_1 , we conduct integration by parts with respect to time as follows:

$$\begin{aligned}
 A_1 &= 2 \int_0^t a_h(\mathbf{u}_I(s) - \mathbf{u}(s), \dot{\mathbf{u}}_I(s) - \dot{\mathbf{u}}_h(s)) \, ds \\
 &= 2 \sum_{T \in \mathcal{T}_h} \int_T \boldsymbol{\sigma}(\mathbf{u}_I(t) - \mathbf{u}(t)) : \boldsymbol{\epsilon}(\mathbf{u}_I(t) - \mathbf{u}_h(t)) \, dx \\
 &\quad - 2 \sum_{T \in \mathcal{T}_h} \int_T \boldsymbol{\sigma}(\mathbf{u}_I(0) - \mathbf{u}(0)) : \boldsymbol{\epsilon}(\mathbf{u}_I(0) - \mathbf{u}_h(0)) \, dx \\
 &\quad - 2 \int_0^t \sum_{T \in \mathcal{T}_h} \int_T \boldsymbol{\sigma}(\dot{\mathbf{u}}_I(s) - \dot{\mathbf{u}}(s)) : \boldsymbol{\epsilon}(\mathbf{u}_I(s) - \mathbf{u}_h(s)) \, dx \, ds \\
 &\lesssim \sum_{T \in \mathcal{T}_h} |\mathbf{u}_I(t) - \mathbf{u}(t)|_{H^1(T)} |\mathbf{u}_I(t) - \mathbf{u}_h(t)|_{H^1(T)} \\
 &\quad + \int_0^t \sum_{T \in \mathcal{T}_h} |\dot{\mathbf{u}}_I(s) - \dot{\mathbf{u}}(s)|_{H^1(T)} |\mathbf{u}_I(s) - \mathbf{u}_h(s)|_{H^1(T)} \, ds,
 \end{aligned}$$

where we have used the fact that $\mathbf{u}_I(0) = \mathbf{u}_h(0)$ in the last step. A use of Young's inequality in the last relation yields

$$\begin{aligned}
 A_1 &\lesssim \frac{1}{\epsilon} \sum_{T \in \mathcal{T}_h} |\mathbf{u}_I(t) - \mathbf{u}(t)|_{H^1(T)}^2 + \epsilon \sum_{T \in \mathcal{T}_h} |\mathbf{u}_I(t) - \mathbf{u}_h(t)|_{H^1(T)}^2 \\
 &\quad + \int_0^t \frac{1}{\epsilon} \sum_{T \in \mathcal{T}_h} |\dot{\mathbf{u}}_I(s) - \dot{\mathbf{u}}(s)|_{H^1(T)}^2 \, ds \\
 &\quad + \int_0^t \epsilon \sum_{T \in \mathcal{T}_h} |\mathbf{u}_I(s) - \mathbf{u}_h(s)|_{H^1(T)}^2 \, ds
 \end{aligned}$$

for some $\epsilon > 0$. Using Lemma 3, we obtain

$$\begin{aligned}
 A_1 &\lesssim \frac{1}{\epsilon} \sum_{T \in \mathcal{T}_h} h_T^2 |\mathbf{u}(t)|_{H^2(T)}^2 + \epsilon \sum_{T \in \mathcal{T}_h} |\mathbf{u}_I(t) - \mathbf{u}_h(t)|_{H^1(T)}^2 + \int_0^t \frac{1}{\epsilon} \sum_{T \in \mathcal{T}_h} h_T^2 |\dot{\mathbf{u}}(s)|_{H^2(T)}^2 \, ds \\
 &\quad + \int_0^t \epsilon \|\mathbf{u}_I(s) - \mathbf{u}_h(s)\|^2 \, ds \\
 &\lesssim \frac{1}{\epsilon} h^2 \|\mathbf{u}\|_{L^\infty(\mathcal{I}; H^2(\Omega))}^2 + \epsilon \|\mathbf{u}_I - \mathbf{u}_h\|_{L^\infty(\mathcal{I}; V_h)}^2 + \frac{1}{\epsilon} T h^2 \|\dot{\mathbf{u}}\|_{L^\infty(\mathcal{I}; H^2(\Omega))}^2 \\
 &\quad + \epsilon T \|\mathbf{u}_I - \mathbf{u}_h\|_{L^\infty(\mathcal{I}; V_h)}^2.
 \end{aligned}$$

Next, we handle term A_2 in the following way:

$$\begin{aligned}
 A_2 &= 2 \int_0^t \sum_{e \in \mathcal{E}_h^C} \int_e \boldsymbol{\sigma}_\tau(\mathbf{u}(s)) \cdot (\dot{\mathbf{u}}_I(s) - \dot{\mathbf{u}}_h(s))_\tau \, d\sigma \, ds + 2 \int_0^t (j(\dot{\mathbf{u}}_I(s)) - j(\dot{\mathbf{u}}_h(s))) \, ds \\
 &\leq 2 \int_0^t \sum_{e \in \mathcal{E}_h^C} \int_e \boldsymbol{\sigma}_\tau(\mathbf{u}(s)) \cdot (\dot{\mathbf{u}}_I(s))_\tau \, d\sigma \, ds + 2 \int_0^t \sum_{e \in \mathcal{E}_h^C} \int_e g_a |(\dot{\mathbf{u}}_h(s))_\tau| \, d\sigma \, ds
 \end{aligned}$$

$$\begin{aligned}
 & + 2 \int_0^t \sum_{e \in \mathcal{E}_h^C} \int_e (g_a |(\mathbf{u}_I(s))_\tau| - g_a |(\mathbf{u}_h(s))_\tau|) \, d\sigma \, ds \\
 & = 2 \int_0^t \sum_{e \in \mathcal{E}_h^C} \int_e \boldsymbol{\sigma}_\tau(\mathbf{u}(s)) \cdot (\mathbf{u}_I(s))_\tau \, d\sigma \, ds + 2 \int_0^t \sum_{e \in \mathcal{E}_h^C} \int_e g_a |(\mathbf{u}_I(s))_\tau| \, d\sigma \, ds.
 \end{aligned}$$

Using the fact that for any contact edge e , we have $\boldsymbol{\sigma}_\tau(\mathbf{u}(t)) \cdot (\dot{\mathbf{u}}(t))_\tau = -g_a |(\dot{\mathbf{u}}(t))_\tau|$, we deduce

$$\begin{aligned}
 A_2 & \leq 2 \int_0^t \sum_{e \in \mathcal{E}_h^C} \int_e \boldsymbol{\sigma}_\tau(\mathbf{u}(s)) \cdot (\mathbf{u}_I(s) - \dot{\mathbf{u}}(s))_\tau \, d\sigma \, ds + 2 \int_0^t \sum_{e \in \mathcal{E}_h^C} \int_e g_a |(\mathbf{u}_I(s))_\tau| \, d\sigma \, ds \\
 & \quad - 2 \int_0^t \sum_{e \in \mathcal{E}_h^C} \int_e g_a |(\dot{\mathbf{u}}(s))_\tau| \, d\sigma \, ds \\
 & \leq 4 \int_0^t \sum_{e \in \mathcal{E}_h^C} \int_e g_a |(\mathbf{u}_I(s))_\tau - (\dot{\mathbf{u}}(s))_\tau| \, d\sigma \, ds \\
 & \lesssim \|g_a\|_{L^\infty(\Gamma_C)} \int_0^t \sum_{e \in \mathcal{E}_h^C} h_e^{\frac{1}{2}} \|\mathbf{u}_I(s) - \dot{\mathbf{u}}(s)\|_{L^2(e)} \, ds.
 \end{aligned}$$

By applying the trace inequality, we obtain

$$A_2 \lesssim \|g_a\|_{L^\infty(\Gamma_C)} \int_0^t \sum_{e \in \mathcal{E}_h^C} \left(\|\mathbf{u}_I(s) - \dot{\mathbf{u}}(s)\|_{L^2(\mathcal{T}_e)} + h_e |\mathbf{u}_I(s) - \dot{\mathbf{u}}(s)|_{H^1(\mathcal{T}_e)} \right) \, ds.$$

Using the commutative property of the interpolation operator $\mathbf{u}_I$ and Lemma 3, we obtain

$$A_2 \lesssim \|g_a\|_{L^\infty(\Gamma_C)} h^2 \|\dot{\mathbf{u}}\|_{L^\infty(\mathcal{I}; H^2(\Omega))}.$$

Now, we will proceed to bound A_3 ,

$$\begin{aligned}
 A_3 & = 2 \int_0^t \sum_{e \in \mathcal{E}_h^0} \int_e \llbracket \boldsymbol{\sigma}(\mathbf{u}(s)) \rrbracket : \llbracket \mathbf{u}_I(s) - \mathbf{u}_h(s) \rrbracket \, d\sigma \, ds \\
 & = 2 \int_0^t \sum_{e \in \mathcal{E}_h^0} \int_e \llbracket \boldsymbol{\sigma}(\mathbf{u}(s) - \mathbf{u}_I(s)) \rrbracket : \llbracket \mathbf{u}_I(s) - \mathbf{u}_h(s) \rrbracket \, d\sigma \, ds \\
 & \quad + 2 \int_0^t \sum_{e \in \mathcal{E}_h^0} \int_e \llbracket \boldsymbol{\sigma}(\mathbf{u}_I(s)) \rrbracket : \llbracket \mathbf{u}_I(s) - \mathbf{u}_h(s) \rrbracket \, d\sigma \, ds. \tag{22}
 \end{aligned}$$

First, we focus on the second term of the last equality. Utilizing that $\boldsymbol{\sigma}(\mathbf{u}_I(t))$ is a constant term on each element, we have

$$2 \int_0^t \sum_{e \in \mathcal{E}_h^0} \int_e \llbracket \boldsymbol{\sigma}(\mathbf{u}_I(s)) \rrbracket : \llbracket \mathbf{u}_I(s) - \mathbf{u}_h(s) \rrbracket \, d\sigma \, ds = \mathbf{0}$$

since $\llbracket \dot{\mathbf{u}}_I(t) - \dot{\mathbf{u}}_h(t) \rrbracket(m_e) = \mathbf{0}$ for all $e \in \mathcal{E}_h^0$. Performing integration by parts with respect to time in the first term of equality (22), we obtain

$$\begin{aligned}
 & 2 \int_0^t \sum_{e \in \mathcal{E}_h^0} \int_e \llbracket \boldsymbol{\sigma}(\mathbf{u}(s) - \mathbf{u}_I(s)) \rrbracket : \llbracket \dot{\mathbf{u}}_I(s) - \dot{\mathbf{u}}_h(s) \rrbracket \, d\sigma \, ds \\
 &= -2 \int_0^t \sum_{e \in \mathcal{E}_h^0} \int_e \llbracket \boldsymbol{\sigma}(\dot{\mathbf{u}}(s) - \dot{\mathbf{u}}_I(s)) \rrbracket : \llbracket \mathbf{u}_I(s) - \mathbf{u}_h(s) \rrbracket \, d\sigma \, ds \\
 &\quad + 2 \sum_{e \in \mathcal{E}_h^0} \int_e \llbracket \boldsymbol{\sigma}(\mathbf{u}(t) - \mathbf{u}_I(t)) \rrbracket : \llbracket \mathbf{u}_I(t) - \mathbf{u}_h(t) \rrbracket \, d\sigma \\
 &\lesssim \int_0^t \sum_{e \in \mathcal{E}_h^0} |\dot{\mathbf{u}}(s) - \dot{\mathbf{u}}_I(s)|_{H^1(e)} \|\llbracket \mathbf{u}_I(s) - \mathbf{u}_h(s) \rrbracket\|_{L^2(e)} \, ds \\
 &\quad + \sum_{e \in \mathcal{E}_h^0} |\mathbf{u}(t) - \mathbf{u}_I(t)|_{H^1(e)} \|\llbracket \mathbf{u}_I(t) - \mathbf{u}_h(t) \rrbracket\|_{L^2(e)}. \tag{23}
 \end{aligned}$$

Employing the trace inequality and properties of the interpolation operator in relation (23), we find

$$\begin{aligned}
 & 2 \int_0^t \sum_{e \in \mathcal{E}_h^0} \int_e \llbracket \boldsymbol{\sigma}(\mathbf{u}(s) - \mathbf{u}_I(s)) \rrbracket : \llbracket \dot{\mathbf{u}}_I(s) - \dot{\mathbf{u}}_h(s) \rrbracket \, d\sigma \, ds \\
 &\lesssim \int_0^t \sum_{e \in \mathcal{E}_h^0} h_e^{\frac{1}{2}} \|\dot{\mathbf{u}}(s)\|_{H^2(\mathcal{T}_e)} \|\llbracket \mathbf{u}_I(s) - \mathbf{u}_h(s) \rrbracket\|_{L^2(e)} \, ds \\
 &\quad + \sum_{e \in \mathcal{E}_h^0} h_e^{\frac{1}{2}} \|\mathbf{u}(t)\|_{H^2(\mathcal{T}_e)} \|\llbracket \mathbf{u}_I(t) - \mathbf{u}_h(t) \rrbracket\|_{L^2(e)}. \tag{24}
 \end{aligned}$$

Next, we estimate $\|\llbracket \mathbf{u}_I(t) - \mathbf{u}_h(t) \rrbracket\|_{L^2(e)}$ as follows. Since $\llbracket \mathbf{u}_I(t) - \mathbf{u}_h(t) \rrbracket$ becomes zero at the midpoint of every edge $e \in \mathcal{E}_h^0$, thus we have $\int_e \llbracket \mathbf{u}_I(t) - \mathbf{u}_h(t) \rrbracket \, ds = 0$ using the midpoint rule. Applying the Poincaré-Wirtinger inequality [19] and trace inequality, we obtain

$$\|\llbracket \mathbf{u}_I(t) - \mathbf{u}_h(t) \rrbracket\|_{L^2(e)} \lesssim h_e \|\llbracket \mathbf{u}_I(t) - \mathbf{u}_h(t) \rrbracket\|_{H^1(e)} \lesssim h_e^{\frac{1}{2}} |\mathbf{u}_I(t) - \mathbf{u}_h(t)|_{H^1(\mathcal{T}_e)}. \tag{25}$$

Inserting Eq. (25) in relation (24), we get

$$\begin{aligned}
 & 2 \int_0^t \sum_{e \in \mathcal{E}_h^0} \int_e \llbracket \boldsymbol{\sigma}(\mathbf{u}(s) - \mathbf{u}_I(s)) \rrbracket : \llbracket \dot{\mathbf{u}}_I(s) - \dot{\mathbf{u}}_h(s) \rrbracket \, d\sigma \, ds \\
 &\lesssim \int_0^t \sum_{e \in \mathcal{E}_h^0} h_e \|\dot{\mathbf{u}}(s)\|_{H^2(\mathcal{T}_e)} |\mathbf{u}_I(s) - \mathbf{u}_h(s)|_{H^1(\mathcal{T}_e)} \, ds \\
 &\quad + \sum_{e \in \mathcal{E}_h^0} h_e \|\mathbf{u}(t)\|_{H^2(\mathcal{T}_e)} |\mathbf{u}_I(t) - \mathbf{u}_h(t)|_{H^1(\mathcal{T}_e)}.
 \end{aligned}$$

Finally, an application of Young's inequality yields

$$2 \int_0^t \sum_{e \in \mathcal{E}_h^0} \int_e \llbracket \boldsymbol{\sigma}(\mathbf{u}(s) - \mathbf{u}_I(s)) \rrbracket : \llbracket \dot{\mathbf{u}}_I(s) - \dot{\mathbf{u}}_h(s) \rrbracket \, d\sigma \, ds$$

$$\begin{aligned}
&\lesssim \int_0^t \left(\frac{1}{\epsilon} \sum_{e \in \mathcal{E}_h^0} h_e^2 \|\dot{\mathbf{u}}(s)\|_{\mathbf{H}^2(\mathcal{T}_e)}^2 \, ds + \epsilon \sum_{e \in \mathcal{E}_h^0} |\mathbf{u}_I(s) - \mathbf{u}_h(s)|_{\mathbf{H}^1(\mathcal{T}_e)}^2 \right) ds \\
&\quad + \frac{1}{\epsilon} \sum_{e \in \mathcal{E}_h^0} h_e^2 \|\mathbf{u}(t)\|_{\mathbf{H}^2(\mathcal{T}_e)}^2 + \epsilon \sum_{e \in \mathcal{E}_h^0} |\mathbf{u}_I(t) - \mathbf{u}_h(t)|_{\mathbf{H}^1(\mathcal{T}_e)}^2 \\
&\lesssim \frac{1}{\epsilon} h_e^2 \|\mathbf{u}\|_{L^\infty(\mathcal{T}; \mathbf{H}^2(\Omega))}^2 + \frac{1}{\epsilon} h_e^2 \|\dot{\mathbf{u}}\|_{L^\infty(\mathcal{T}; \mathbf{H}^2(\Omega))}^2 \\
&\quad + \epsilon \|\mathbf{u}_I - \mathbf{u}_h\|_{L^\infty(\mathcal{T}; \mathbf{V}_h)}^2.
\end{aligned}$$

The bound on A_4 can be derived as follows:

$$\begin{aligned}
A_4 &= 2 \int_0^t \sum_{e \in \mathcal{E}_h^C} \int_e \sigma_n(\mathbf{u}(s)) \cdot (\dot{\mathbf{u}}_I(s) - \dot{\mathbf{u}}_h(s))_n \, d\sigma \, ds \\
&= 2 \int_0^t \sum_{e \in \mathcal{E}_h^C} \int_e \sigma_n(\mathbf{u}(s) - \mathbf{u}_I(s)) \cdot (\dot{\mathbf{u}}_I(s) - \dot{\mathbf{u}}_h(s))_n \, d\sigma \, ds \\
&\quad + 2 \int_0^t \sum_{e \in \mathcal{E}_h^C} \int_e \sigma_n(\mathbf{u}_I(s)) \cdot (\dot{\mathbf{u}}_I(s) - \dot{\mathbf{u}}_h(s))_n \, d\sigma \, ds.
\end{aligned}$$

The last term in the above relation vanishes as for any contact edge $e \in \mathcal{E}_h^C$, we have $\sigma_n(\mathbf{u}_I(s))$ is constant and $\int_e (\dot{\mathbf{u}}_I(s) - \dot{\mathbf{u}}_h(s))_n \, d\sigma = 0$. Thus, we obtain

$$A_4 = 2 \int_0^t \sum_{e \in \mathcal{E}_h^C} \int_e \sigma_n(\mathbf{u}(s) - \mathbf{u}_I(s)) \cdot (\dot{\mathbf{u}}_I(s) - \dot{\mathbf{u}}_h(s))_n \, d\sigma \, ds. \quad (26)$$

Next, the bound of Eq. (26) can be derived analogously using similar arguments as in the case of A_3 . Consequently, we obtain the following estimate:

$$\begin{aligned}
A_4 &\lesssim \frac{1}{\epsilon} h_e^2 \|\mathbf{u}\|_{L^\infty(\mathcal{T}; \mathbf{H}^2(\Omega))}^2 + \frac{1}{\epsilon} h_e^2 \|\dot{\mathbf{u}}\|_{L^\infty(\mathcal{T}; \mathbf{H}^2(\Omega))}^2 \\
&\quad + \epsilon \|\mathbf{u}_I - \mathbf{u}_h\|_{L^\infty(\mathcal{T}; \mathbf{V}_h)}^2.
\end{aligned}$$

Therefore, by substituting the bounds obtained for A_1 , A_2 , A_3 , and A_4 in relation (21) and choosing $\epsilon > 0$ sufficiently small, we obtain

$$\|\mathbf{u}_I - \mathbf{u}_h\|_{L^\infty(\mathcal{T}; \mathbf{V}_h)}^2 \lesssim h^2.$$

Finally, combining the upper bound of $\|\boldsymbol{\theta}\|_{L^\infty(\mathcal{T}; \mathbf{V}_h)}$, we arrive at the final result.

4 Fully Discrete Approximation

In addition to divide the spatial domain into the regular triangulation, we also partition the time interval $[0, T]$ into subintervals $[t_{n-1}, t_n]$ for $n = 1, 2, \dots, N$, where $0 = t_0 < t_1 < t_2 < t_3 < \dots < t_N = T$. We represent the length of the subinterval $[t_{n-1}, t_n]$ as k_n with k denoting the maximum value among all k_n . At each time step t_n , the fully discrete solution is identified by $\mathbf{u}^n$. For the linear functionals $\mathbf{f}$ and $\mathbf{g}$ we use the notations $\mathbf{f}^n = \mathbf{f}(t_n)$ and $\mathbf{g}^n = \mathbf{g}(t_n)$ to describe their values at the time level t_n . It is important to note that the above notations are well defined due to the Sobolev imbedding theorem, which guarantees

that $W^{1,\infty}(\mathcal{I}; X) \hookrightarrow C(\mathcal{I}; X)$ for any Banach space X . We have employed the backward divided difference scheme to approximate the derivatives, i.e., $\delta \mathbf{u}^n = \frac{\mathbf{u}^n - \mathbf{u}^{n-1}}{k_n}$.

Using the backward Euler scheme, we obtain the following fully discrete approximation of the variational inequality (10).

Problem 2 (Fully Discrete Variational Inequality) For each $n = 1, 2, \dots, N$, find $\{\mathbf{u}_h^n\}_{n=0}^N \in \mathcal{K}_{CR}^h$ such that

$$a_h(\mathbf{u}_h^n, \mathbf{v}_h - \delta \mathbf{u}_h^n) + j(\mathbf{v}_h) - j(\delta \mathbf{u}_h^n) \geq (\mathbf{l}^n, \mathbf{v}_h - \delta \mathbf{u}_h^n), \quad \forall \mathbf{v}_h \in \mathcal{K}_{CR}^h, \quad (27)$$

$$\mathbf{u}_h^0 = \mathbf{I}^{CR} \mathbf{u}_0. \quad (28)$$

The existence and uniqueness of the solution of Problem 2 are guaranteed by the Lions-Stampacchia theorem [17].

4.1 Convergence Analysis for Fully Discrete Approximation

In this subsection, we will derive a priori error estimates for the fully discrete scheme.

Theorem 2 Let $\mathbf{u}$ be the solution of the continuous problem (10)–(11) and $\partial \{\mathbf{u}_h^n\}_n = 0^N$ be the solution of the fully discrete problem (27)–(28). Assume $\mathbf{u}, \dot{\mathbf{u}} \in L^\infty[\mathcal{I}; \mathbf{H}^2(\Omega)]$ and $\ddot{\mathbf{u}} \in L^1[\mathcal{I}; \mathbf{H}^2(\Omega)]$. Then, we have

$$\max_n \|\mathbf{u}^n - \mathbf{u}_h^n\| \lesssim h + k.$$

Proof Let $\mathbf{e}^n = \mathbf{u}^n - \mathbf{u}_h^n$. Decompose the error into $\mathbf{e}^n = \boldsymbol{\theta}^n + \boldsymbol{\eta}^n$ where $\boldsymbol{\eta}^n = \mathbf{u}^n - \mathbf{u}_I^n$ and $\boldsymbol{\theta}^n = \mathbf{u}_I^n - \mathbf{u}_h^n$. Taking into account the continuity of the bilinear form $a_h(\cdot, \cdot)$ with respect to the norm $\|\cdot\|$, we first obtain the lower bound of $a_h(\mathbf{e}^n, \delta \mathbf{e}^n)$ as follows:

$$\begin{aligned} a_h(\mathbf{e}^n, \delta \mathbf{e}^n) &= \frac{1}{k_n} \left(a_h(\mathbf{e}^n, \mathbf{e}^n) - a_h(\mathbf{e}^n, \mathbf{e}^{n-1}) \right) \\ &\geq \frac{1}{k_n} \left(\|\mathbf{e}^n\|^2 - \|\mathbf{e}^n\| \|\mathbf{e}^{n-1}\| \right) \\ &\geq \frac{1}{2k_n} \left(\|\mathbf{e}^n\|^2 - \|\mathbf{e}^{n-1}\|^2 \right). \end{aligned}$$

Next, we determine the upper bound for $a_h(\mathbf{e}^n, \delta \mathbf{e}^n)$ in the following manner:

$$\begin{aligned} a_h(\mathbf{e}^n, \delta \mathbf{e}^n) &= a_h(\mathbf{e}^n, \delta \boldsymbol{\theta}^n) + a_h(\mathbf{e}^n, \delta \boldsymbol{\eta}^n) \\ &= a_h(\mathbf{e}^n, \delta \boldsymbol{\eta}^n) + a_h(\mathbf{u}^n, \delta \mathbf{u}_I^n - \delta \mathbf{u}_h^n) - a_h(\mathbf{u}_h^n, \delta \mathbf{u}_I^n - \delta \mathbf{u}_h^n). \end{aligned} \quad (29)$$

Since $\delta \mathbf{u}_I^n \in \mathcal{K}_{CR}^h$, using Eq. (27), we find

$$-a_h(\mathbf{u}_h^n, \delta \mathbf{u}_I^n - \delta \mathbf{u}_h^n) \leq j(\delta \mathbf{u}_I^n) - j(\delta \mathbf{u}_h^n) - (\mathbf{l}^n, \delta \mathbf{u}_I^n - \delta \mathbf{u}_h^n). \quad (30)$$

Inserting Eq. (30) in relation (29), we get

$$\begin{aligned} a_h(\mathbf{e}^n, \delta \mathbf{e}^n) &\leq a_h(\mathbf{e}^n, \delta \boldsymbol{\eta}^n) + a_h(\mathbf{u}^n, \delta \mathbf{u}_I^n - \delta \mathbf{u}_h^n) \\ &\quad + j(\delta \mathbf{u}_I^n) - j(\delta \mathbf{u}_h^n) - (\mathbf{l}^n, \delta \mathbf{u}_I^n - \delta \mathbf{u}_h^n). \end{aligned} \quad (31)$$

Owing to integration by parts in the second term of the above inequality, we get

$$a_h(\mathbf{u}^n, \delta \mathbf{u}_I^n - \delta \mathbf{u}_h^n) = \sum_{T \in \mathcal{T}_h} \int_T \boldsymbol{\sigma}(\mathbf{u}^n) : \boldsymbol{\epsilon}(\delta \mathbf{u}_I^n - \delta \mathbf{u}_h^n) \, dx$$

$$\begin{aligned}
&= \sum_{T \in \mathcal{T}_h} \int_T f(t_n) \cdot (\delta \mathbf{u}_I^n - \delta \mathbf{u}_h^n) \, dx + \sum_{e \in \mathcal{E}_h^N} \int_e \mathbf{g}(t_n) \cdot (\delta \mathbf{u}_I^n - \delta \mathbf{u}_h^n) \, d\sigma \\
&\quad + \sum_{e \in \mathcal{E}_h^C} \int_e (\boldsymbol{\sigma}(\mathbf{u}^n) \mathbf{n}) \cdot (\delta \mathbf{u}_I^n - \delta \mathbf{u}_h^n) \, d\sigma + \sum_{e \in \mathcal{E}_h^0} \int_e \{\{\boldsymbol{\sigma}(\mathbf{u}^n)\}\} : \llbracket \delta \mathbf{u}_I^n - \delta \mathbf{u}_h^n \rrbracket \, d\sigma. \quad (32)
\end{aligned}$$

Combining Eqs. (31) and (32), we obtain

$$\begin{aligned}
a_h(\mathbf{e}^n, \delta \mathbf{e}^n) &\leq a_h(\mathbf{e}^n, \delta \boldsymbol{\eta}^n) + \sum_{e \in \mathcal{E}_h^0} \int_e \{\{\boldsymbol{\sigma}(\mathbf{u}^n)\}\} : \llbracket \delta \mathbf{u}_I^n - \delta \mathbf{u}_h^n \rrbracket \, d\sigma + j(\delta \mathbf{u}_I^n) - j(\delta \mathbf{u}_h^n) \\
&\quad + \sum_{e \in \mathcal{E}_h^C} \int_e (\boldsymbol{\sigma}_n(\mathbf{u}^n))(\delta \mathbf{u}_I^n - \delta \mathbf{u}_h^n)_n \, d\sigma + \sum_{e \in \mathcal{E}_h^C} \int_e (\boldsymbol{\sigma}_\tau(\mathbf{u}^n)) \cdot (\delta \mathbf{u}_I^n - \delta \mathbf{u}_h^n)_\tau \, d\sigma \\
&= T_1 + T_2 + T_3 + T_4, \quad (33)
\end{aligned}$$

where

$$\begin{aligned}
T_1 &:= \sum_{e \in \mathcal{E}_h^0} \int_e \{\{\boldsymbol{\sigma}(\mathbf{u}^n)\}\} : \llbracket \delta \mathbf{u}_I^n - \delta \mathbf{u}_h^n \rrbracket \, d\sigma, \\
T_2 &:= j(\delta \mathbf{u}_I^n) - j(\delta \mathbf{u}_h^n) + \sum_{e \in \mathcal{E}_h^C} \int_e (\boldsymbol{\sigma}_\tau(\mathbf{u}^n)) \cdot (\delta \mathbf{u}_I^n - \delta \mathbf{u}_h^n)_\tau \, d\sigma, \\
T_3 &:= a_h(\mathbf{e}^n, \delta \boldsymbol{\eta}^n), \\
T_4 &:= \sum_{e \in \mathcal{E}_h^C} \int_e (\boldsymbol{\sigma}_n(\mathbf{u}^n))(\delta \mathbf{u}_I^n - \delta \mathbf{u}_h^n)_n \, d\sigma.
\end{aligned}$$

• *Estimates for T_1 :*

$$\begin{aligned}
T_1 &= \sum_{e \in \mathcal{E}_h^0} \int_e \{\{\boldsymbol{\sigma}(\mathbf{u}^n)\}\} : \llbracket \delta \mathbf{u}_I^n - \delta \mathbf{u}_h^n \rrbracket \, d\sigma \\
&= \sum_{e \in \mathcal{E}_h^0} \int_e \{\{\boldsymbol{\sigma}(\mathbf{u}^n - \mathbf{u}_I^n)\}\} : \llbracket \delta \mathbf{u}_I^n - \delta \mathbf{u}_h^n \rrbracket \, d\sigma + \sum_{e \in \mathcal{E}_h^0} \int_e \{\{\boldsymbol{\sigma}(\mathbf{u}_I^n)\}\} : \llbracket \delta \mathbf{u}_I^n - \delta \mathbf{u}_h^n \rrbracket \, d\sigma.
\end{aligned}$$

Observe that the second term in the last relation vanishes as $\{\{\boldsymbol{\sigma}(\mathbf{u}_I^n)\}\}$ is constant and $\sum_{e \in \mathcal{E}_h^0} \int_e \llbracket \delta \mathbf{u}_I^n - \delta \mathbf{u}_h^n \rrbracket \, d\sigma = 0$ on any edge $e \in \mathcal{E}_h^0$. Thus, T_1 reduces to

$$\begin{aligned}
T_1 &= \sum_{e \in \mathcal{E}_h^0} \int_e \{\{\boldsymbol{\sigma}(\mathbf{u}^n - \mathbf{u}_I^n)\}\} : \llbracket \delta \mathbf{u}_I^n - \delta \mathbf{u}_h^n \rrbracket \, d\sigma \\
&= \sum_{e \in \mathcal{E}_h^0} \int_e \{\{\boldsymbol{\sigma}(\mathbf{u}^n - \mathbf{u}_I^n)\}\} : \llbracket \delta \mathbf{u}_I^n - \delta \mathbf{u}^n \rrbracket \, d\sigma + \sum_{e \in \mathcal{E}_h^0} \int_e \{\{\boldsymbol{\sigma}(\mathbf{u}^n - \mathbf{u}_I^n)\}\} : \llbracket \delta \mathbf{u}^n - \delta \mathbf{u}_h^n \rrbracket \, d\sigma \\
&= \sum_{e \in \mathcal{E}_h^0} \int_e \{\{\boldsymbol{\sigma}(\mathbf{u}^n - \mathbf{u}_I^n)\}\} : \llbracket \delta \mathbf{u}_I^n \rrbracket \, d\sigma + \sum_{e \in \mathcal{E}_h^0} \int_e \{\{\boldsymbol{\sigma}(\mathbf{u}^n - \mathbf{u}_I^n)\}\} : \llbracket \delta \mathbf{u}^n - \delta \mathbf{u}_h^n \rrbracket \, d\sigma \\
&= \sum_{e \in \mathcal{E}_h^0} \int_e \{\{\boldsymbol{\sigma}(\mathbf{u}^n - \mathbf{u}_I^n)\}\} : \llbracket \delta \mathbf{u}_I^n - \dot{\mathbf{u}}^n \rrbracket \, d\sigma + \sum_{e \in \mathcal{E}_h^0} \int_e \{\{\boldsymbol{\sigma}(\mathbf{u}^n - \mathbf{u}_I^n)\}\} : \llbracket \delta \mathbf{u}^n - \delta \mathbf{u}_h^n \rrbracket \, d\sigma. \quad (34)
\end{aligned}$$

Observe that

$$\delta \mathbf{u}_I^n = \frac{\mathbf{u}_I^n - \mathbf{u}_I^{n-1}}{k_n} = \frac{\mathbf{u}_I^n - (\mathbf{u}_I^n - k_n \dot{\mathbf{u}}_I^n)}{k_n} = \dot{\mathbf{u}}_I^n.$$

A use of the trace inequality and properties of the interpolation map in Eq. (34) yields

$$\begin{aligned} T_1 &\leq \sum_{e \in \mathcal{E}_h^0} \int_e \llbracket \sigma(\mathbf{u}^n - \mathbf{u}_I^n) \rrbracket : \llbracket \delta \mathbf{u}^n - \delta \mathbf{u}_h^n \rrbracket \, d\sigma + \sum_{e \in \mathcal{E}_h^0} |\mathbf{u}^n - \mathbf{u}_I^n|_{H^1(e)} \|\dot{\mathbf{u}}_I^n - \dot{\mathbf{u}}^n\|_{L^2(e)} \\ &\lesssim \sum_{e \in \mathcal{E}_h^0} \int_e \llbracket \sigma(\mathbf{u}^n - \mathbf{u}_I^n) \rrbracket : \llbracket \delta \mathbf{u}^n - \delta \mathbf{u}_h^n \rrbracket \, d\sigma + h^2 \|\mathbf{u}^n\|_{H^2(\Omega)} \|\dot{\mathbf{u}}^n\|_{H^2(\Omega)} \\ &\lesssim \sum_{e \in \mathcal{E}_h^0} \int_e \llbracket \sigma(\mathbf{u}^n - \mathbf{u}_I^n) \rrbracket : \llbracket \delta \mathbf{u}^n - \delta \mathbf{u}_h^n \rrbracket \, d\sigma + h^2 \|\mathbf{u}^n\|_{H^2(\Omega)}^2 + h^2 \|\dot{\mathbf{u}}^n\|_{H^2(\Omega)}^2. \quad (35) \end{aligned}$$

• *Estimates for T_2 :*

$$\begin{aligned} T_2 &= \sum_{e \in \mathcal{E}_h^C} \int_e (\sigma_\tau(\mathbf{u}^n)) \cdot (\delta \mathbf{u}_I^n - \delta \mathbf{u}_h^n)_\tau \, d\sigma + j(\delta \mathbf{u}_I^n) - j(\delta \mathbf{u}_h^n) \\ &\leq \sum_{e \in \mathcal{E}_h^C} \int_e (\sigma_\tau(\mathbf{u}^n)) \cdot (\delta \mathbf{u}_I^n)_\tau \, d\sigma + \sum_{e \in \mathcal{E}_h^C} \int_e g_a |(\delta \mathbf{u}_h^n)_\tau| \, d\sigma \\ &\quad + \sum_{e \in \mathcal{E}_h^C} \int_e g_a |(\delta \mathbf{u}_I^n)_\tau| \, d\sigma - \sum_{e \in \mathcal{E}_h^C} \int_e g_a |(\delta \mathbf{u}_h^n)_\tau| \, d\sigma \\ &= \sum_{e \in \mathcal{E}_h^C} \int_e (\sigma_\tau(\mathbf{u}^n)) \cdot (\delta \mathbf{u}_I^n)_\tau \, d\sigma + \sum_{e \in \mathcal{E}_h^C} \int_e g_a |(\delta \mathbf{u}_I^n)_\tau| \, d\sigma. \quad (36) \end{aligned}$$

Since $\delta \mathbf{u}_I^n = \dot{\mathbf{u}}_I^n$, we have

$$\begin{aligned} T_2 &\leq \sum_{e \in \mathcal{E}_h^C} \int_e (\sigma_\tau(\mathbf{u}^n)) \cdot (\dot{\mathbf{u}}_I^n - \dot{\mathbf{u}}^n)_\tau \, d\sigma \\ &\quad + \sum_{e \in \mathcal{E}_h^C} \int_e (\sigma_\tau(\mathbf{u}^n)) \cdot (\dot{\mathbf{u}}^n)_\tau \, d\sigma + \sum_{e \in \mathcal{E}_h^C} \int_e g_a |(\delta \mathbf{u}_I^n)_\tau| \, d\sigma \end{aligned}$$

Exploiting the fact that $\sigma_\tau(\mathbf{u}^n) \cdot \dot{\mathbf{u}}_\tau^n = -g_a |\dot{\mathbf{u}}_\tau^n|$, the last relation reduces to

$$T_2 \leq 2 \sum_{e \in \mathcal{E}_h^C} \int_e g_a |(\dot{\mathbf{u}}_I^n - \dot{\mathbf{u}}^n)_\tau| \, d\sigma \leq 2 \|g_a\|_{L^\infty(\Gamma_C)} \sum_{e \in \mathcal{E}_h^C} \int_e |(\dot{\mathbf{u}}_I^n - \dot{\mathbf{u}}^n)_\tau| \, d\sigma.$$

Using the Cauchy-Schwarz inequality followed by the trace inequality and Lemma 3, we arrive at

$$\begin{aligned} T_2 &\lesssim \|g_a\|_{L^\infty(\Gamma_C)} \sum_{e \in \mathcal{E}_h^C} h_e^{\frac{1}{2}} \|(\dot{\mathbf{u}}_I^n - \dot{\mathbf{u}}^n)_\tau\|_{L^2(e)} \\ &\lesssim \|g_a\|_{L^\infty(\Gamma_C)} \sum_{e \in \mathcal{E}_h^C} \sum_{T \in \mathcal{T}_e} \left(\|\dot{\mathbf{u}}_I^n - \dot{\mathbf{u}}^n\|_{L^2(T)} + h_e |\dot{\mathbf{u}}_I^n - \dot{\mathbf{u}}^n|_{H^1(T)} \right) \end{aligned}$$

$$\lesssim \|g_a\|_{L^\infty(\Gamma_C)} h^2 |\dot{\mathbf{u}}^n|_{\mathbf{H}^2(\Omega)}.$$

• *Estimates for T_3 :*

$$T_3 = a_h(\mathbf{e}^n, \delta \boldsymbol{\eta}^n) = \frac{1}{k_n} a_h(\mathbf{e}^n, \boldsymbol{\eta}^n - \boldsymbol{\eta}^{n-1}) \leq \frac{1}{k_n} \|\|\|\mathbf{e}^n\|\|\| \|\|\boldsymbol{\eta}^n - \boldsymbol{\eta}^{n-1}\|\|\|.$$

It suffices to bound $\|\|\boldsymbol{\eta}^n - \boldsymbol{\eta}^{n-1}\|\|\|$ as follows:

$$\begin{aligned} \|\|\boldsymbol{\eta}^n - \boldsymbol{\eta}^{n-1}\|\|\| &= \|\|\mathbf{u}^n - \mathbf{u}_I^n - (\mathbf{u}^{n-1} - \mathbf{u}_I^{n-1})\|\|\| \\ &= \|\|\mathbf{u}^n - \mathbf{u}^{n-1} - (\mathbf{u}_I^n - \mathbf{u}_I^{n-1})\|\|\|. \end{aligned}$$

Using Taylor's theorem, we obtain

$$\begin{aligned} \|\|\boldsymbol{\eta}^n - \boldsymbol{\eta}^{n-1}\|\|\| &= \left\| \left\| k_n \dot{\mathbf{u}}_I^{n-1} - k_n \dot{\mathbf{u}}^{n-1} - \int_{t_{n-1}}^{t_n} (t_n - s) \ddot{\mathbf{u}}(s) \, ds \right\| \right\| \\ &\leq \left\| \left\| k_n \dot{\mathbf{u}}_I^{n-1} - k_n \dot{\mathbf{u}}^{n-1} \right\| + \left\| \int_{t_{n-1}}^{t_n} (t_n - s) \ddot{\mathbf{u}}(s) \, ds \right\| \right\| \\ &\leq h k_n |\dot{\mathbf{u}}^{n-1}|_{\mathbf{H}^2(\Omega)} + k_n \left\| \int_{t_{n-1}}^{t_n} \ddot{\mathbf{u}}(s) \, ds \right\|. \end{aligned} \quad (37)$$

• *Estimates for T_4 :*

$$\begin{aligned} T_4 &:= \sum_{e \in \mathcal{E}_h^C} \int_e (\sigma_n(\mathbf{u}^n)) (\delta \mathbf{u}_I^n - \delta \mathbf{u}_h^n)_n \, d\sigma \\ &= \sum_{e \in \mathcal{E}_h^C} \int_e (\sigma_n(\mathbf{u}^n - \mathbf{u}_I^n)) (\delta \mathbf{u}_I^n - \delta \mathbf{u}_h^n)_n \, d\sigma, \end{aligned}$$

since $(\delta \mathbf{u}_I^n - \delta \mathbf{u}_h^n)_n(m_e) = 0$ for any contact edge e . Combining the estimates for T_1 , T_2 , T_3 , and T_4 , we obtain

$$\begin{aligned} a_h(\mathbf{e}^n, \delta \mathbf{e}^n) &\lesssim \sum_{e \in \mathcal{E}_h^0} \int_e \{\|\sigma(\mathbf{u}^n - \mathbf{u}_I^n)\|\} : \|\delta \mathbf{u}^n - \delta \mathbf{u}_h^n\| \, d\sigma \\ &\quad + \sum_{e \in \mathcal{E}_h^C} \int_e (\sigma_n(\mathbf{u}^n - \mathbf{u}_I^n)) (\delta \mathbf{u}_I^n - \delta \mathbf{u}_h^n)_n \, d\sigma \\ &\quad + h^2 \|\mathbf{u}^n\|_{\mathbf{H}^2(\Omega)}^2 + h^2 \|\dot{\mathbf{u}}^n\|_{\mathbf{H}^2(\Omega)}^2 + \|g_a\|_{L^\infty(\Gamma_C)} h^2 |\dot{\mathbf{u}}^n|_{\mathbf{H}^2(\Omega)} \\ &\quad + \frac{1}{k_n} \left(\|\|\mathbf{e}^n\|\|\| (h k_n |\dot{\mathbf{u}}^{n-1}|_{\mathbf{H}^2(\Omega)} + k_n \int_{t_{n-1}}^{t_n} \|\ddot{\mathbf{u}}(s)\| \, ds) \right). \end{aligned}$$

Exploiting the lower bound of $a_h(\mathbf{e}^n, \delta \mathbf{e}^n)$, we deduce

$$\begin{aligned} \frac{1}{2k_n} \left(\|\|\mathbf{e}^n\|\|^2 - \|\|\mathbf{e}^{n-1}\|\|^2 \right) &\lesssim \sum_{e \in \mathcal{E}_h^0} \int_e \{\|\sigma(\mathbf{u}^n - \mathbf{u}_I^n)\|\} : \|\delta \mathbf{u}^n - \delta \mathbf{u}_h^n\| \, d\sigma \\ &\quad + \sum_{e \in \mathcal{E}_h^C} \int_e (\sigma_n(\mathbf{u}^n - \mathbf{u}_I^n)) (\delta \mathbf{u}_I^n - \delta \mathbf{u}_h^n)_n \, d\sigma \\ &\quad + h^2 \|\mathbf{u}^n\|_{\mathbf{H}^2(\Omega)}^2 + h^2 \|\dot{\mathbf{u}}^n\|_{\mathbf{H}^2(\Omega)}^2 + \|g_a\|_{L^\infty(\Gamma_C)} h^2 |\dot{\mathbf{u}}^n|_{\mathbf{H}^2(\Omega)} \end{aligned}$$

$$+ \frac{1}{k_n} \left(\|e^n\| \|hk_n |\dot{u}^{n-1}|_{H^2(\Omega)} + k_n \int_{t_{n-1}}^{t_n} \|\ddot{u}(s)\| \, ds \right).$$

This implies

$$\begin{aligned} \|e^n\|^2 - \|e^{n-1}\|^2 &\lesssim \sum_{e \in \mathcal{E}_h^0} \int_e k_n \langle \sigma(u^n - u_I^n) \rangle : \langle \delta u^n - \delta u_h^n \rangle \, d\sigma \\ &\quad + \sum_{e \in \mathcal{E}_h^C} \int_e k_n (\sigma_n(u^n - u_I^n)) (\delta u_I^n - \delta u_h^n)_n \, d\sigma \\ &\quad + k_n (h^2 \|u^n\|_{H^2(\Omega)}^2 + h^2 \|\dot{u}^n\|_{H^2(\Omega)}^2) \\ &\quad + \|g_a\|_{L^\infty(\Gamma_C)} h^2 k_n |\dot{u}^n|_{H^2(\Omega)} + h k_n \|e^n\| \|\dot{u}^{n-1}\|_{H^2(\Omega)} \\ &\quad + k_n \|e^n\| \int_{t_{n-1}}^{t_n} \|\ddot{u}(s)\| \, ds. \end{aligned}$$

A simple induction from $n = 1, 2, \dots, N$ yields

$$\begin{aligned} \|e^N\|^2 &\lesssim \sum_{j=1}^N \left(\sum_{e \in \mathcal{E}_h^0} \int_e k_j \langle \sigma(u^j - u_I^j) \rangle : \langle \delta u^j - \delta u_h^j \rangle \, d\sigma \right. \\ &\quad \left. + \sum_{e \in \mathcal{E}_h^C} \int_e k_j (\sigma_n(u^j - u_I^j)) (\delta u_I^j - \delta u_h^j)_n \, d\sigma \right) \\ &\quad + \left(h^2 \max_n \|u^n\|_{H^2(\Omega)}^2 + h^2 \max_n \|\dot{u}^n\|_{H^2(\Omega)}^2 \right) \sum_{j=1}^N k_j + \|g_a\|_{L^\infty(\Gamma_C)} h^2 \sum_{j=1}^N k_j \max_n \|\dot{u}^n\|_{H^2(\Omega)} \\ &\quad + h \max_n \|e^n\| \max_n \|\dot{u}^{n-1}\|_{H^2(\Omega)} \sum_{j=1}^N k_j + \max_n \|e^n\| \sum_{j=1}^N k_j \int_{t_{j-1}}^{t_j} \|\ddot{u}(s)\| \, ds. \quad (38) \end{aligned}$$

Next, we will estimate the first term in the above relation as follows:

$$\begin{aligned} &\sum_{j=1}^N \sum_{e \in \mathcal{E}_h^0} \int_e k_j \langle \sigma(u^j - u_I^j) \rangle : \langle \delta u^j - \delta u_h^j \rangle \, d\sigma \\ &= \sum_{j=1}^N \sum_{e \in \mathcal{E}_h^0} \int_e \langle \sigma(u^j - u_I^j) \rangle : \langle (u^j - u^{j-1}) - (u_h^j - u_h^{j-1}) \rangle \, d\sigma \\ &= \sum_{j=1}^N \sum_{e \in \mathcal{E}_h^0} \int_e \langle (u^j - u_h^j) - (u^{j-1} - u_h^{j-1}) \rangle : \langle \sigma(u^j - u_I^j) \rangle \, d\sigma. \end{aligned}$$

Upon re-arranging these terms, we obtain

$$\begin{aligned} &\sum_{j=1}^N \sum_{e \in \mathcal{E}_h^0} \int_e \langle (u^j - u_h^j) - (u^{j-1} - u_h^{j-1}) \rangle : \langle \sigma(u^j - u_I^j) \rangle \, d\sigma \\ &= \sum_{e \in \mathcal{E}_h^0} \int_e \langle u^N - u_h^N \rangle : \langle \sigma(u^N - u_I^N) \rangle \, d\sigma - \sum_{e \in \mathcal{E}_h^0} \int_e \langle u^0 - u_h^0 \rangle : \langle \sigma(u^1 - u_I^1) \rangle \, d\sigma \end{aligned}$$

$$+ \sum_{j=1}^{N-1} \sum_{e \in \mathcal{E}_h^0} \int_e \llbracket \mathbf{u}^j - \mathbf{u}_h^j \rrbracket : \llbracket \sigma(\mathbf{u}^j - \mathbf{u}_I^j) - \sigma(\mathbf{u}^{j+1} - \mathbf{u}_I^{j+1}) \rrbracket \, d\sigma.$$

Using the Cauchy-Schwarz inequality, we find

$$\begin{aligned} & \sum_{j=1}^N \sum_{e \in \mathcal{E}_h^0} \int_e \llbracket (\mathbf{u}^j - \mathbf{u}_h^j) - (\mathbf{u}^{j-1} - \mathbf{u}_h^{j-1}) \rrbracket : \llbracket \sigma(\mathbf{u}^j - \mathbf{u}_I^j) \rrbracket \, d\sigma \\ &= \sum_{e \in \mathcal{E}_h^0} \|\llbracket \mathbf{u}^N - \mathbf{u}_h^N \rrbracket\|_{L^2(e)} \|\llbracket \sigma(\mathbf{u}^N - \mathbf{u}_I^N) \rrbracket\|_{L^2(e)} \\ &+ \sum_{e \in \mathcal{E}_h^0} \|\llbracket \mathbf{u}^0 - \mathbf{u}_h^0 \rrbracket\|_{L^2(e)} \|\llbracket \sigma(\mathbf{u}^1 - \mathbf{u}_I^1) \rrbracket\|_{L^2(e)} \\ &+ \sum_{j=1}^{N-1} \sum_{e \in \mathcal{E}_h^0} \|\llbracket \mathbf{u}^j - \mathbf{u}_h^j \rrbracket\|_{L^2(e)} \|\llbracket \sigma(\mathbf{u}^j - \mathbf{u}_I^j) - \sigma(\mathbf{u}^{j+1} - \mathbf{u}_I^{j+1}) \rrbracket\|_{L^2(e)} \\ &\lesssim \|e^N\|_h \|\mathbf{u}^N\|_{H^2(\Omega)} + \|e^0\|_h \|\mathbf{u}^1\|_{H^2(\Omega)} \\ &+ \sum_{j=1}^{N-1} \sum_{e \in \mathcal{E}_h^0} \|\llbracket \mathbf{u}^j - \mathbf{u}_h^j \rrbracket\|_{L^2(e)} \|\llbracket \sigma(\mathbf{u}^j - \mathbf{u}_I^j) - \sigma(\mathbf{u}^{j+1} - \mathbf{u}_I^{j+1}) \rrbracket\|_{L^2(e)} \end{aligned}$$

as a result of the trace inequality and properties of the interpolation map. Now, consider

$$\begin{aligned} & \sum_{j=1}^{N-1} \sum_{e \in \mathcal{E}_h^0} \|\llbracket \mathbf{u}^j - \mathbf{u}_h^j \rrbracket\|_{L^2(e)} \|\llbracket \sigma(\mathbf{u}^j - \mathbf{u}_I^j) - \sigma(\mathbf{u}^{j+1} - \mathbf{u}_I^{j+1}) \rrbracket\|_{L^2(e)} \\ &\leq \sum_{j=1}^{N-1} \left(\sum_{e \in \mathcal{E}_h^0} h_e^{-1} \|\llbracket \mathbf{u}^j - \mathbf{u}_h^j \rrbracket\|_{L^2(e)}^2 \right)^{\frac{1}{2}} \left(\sum_{e \in \mathcal{E}_h^0} h_e \|\llbracket \sigma(\mathbf{u}^j - \mathbf{u}_I^j) - \sigma(\mathbf{u}^{j+1} - \mathbf{u}_I^{j+1}) \rrbracket\|_{L^2(e)}^2 \right)^{\frac{1}{2}} \\ &\lesssim \sum_{j=1}^{N-1} \|e^j\| \left(\sum_{e \in \mathcal{E}_h^0} h_e \|\llbracket \sigma(\mathbf{u}^j - \mathbf{u}_I^j) - \sigma(\mathbf{u}^{j+1} - \mathbf{u}_I^{j+1}) \rrbracket\|_{L^2(e)}^2 \right)^{\frac{1}{2}} \\ &\lesssim \sum_{j=1}^{N-1} \|e^j\| \left(\sum_{e \in \mathcal{E}_h^0} h_e |(\mathbf{u}^j - \mathbf{u}_I^j) - (\mathbf{u}^{j+1} - \mathbf{u}_I^{j+1})|_{H^1(e)}^2 \right)^{\frac{1}{2}} \\ &= \sum_{j=1}^{N-1} \|e^j\| \left(\sum_{e \in \mathcal{E}_h^0} h_e k_{j+1}^2 |\delta \boldsymbol{\eta}^{j+1}|_{H^1(e)}^2 \right)^{\frac{1}{2}}. \end{aligned} \quad (39)$$

For any $j = 1, 2, \dots, N-1$, using the trace inequality, we find

$$\begin{aligned} |\delta \boldsymbol{\eta}^{j+1}|_{H^1(e)} &\lesssim h_e^{-\frac{1}{2}} |\delta \boldsymbol{\eta}^{j+1}|_{H^1(\mathcal{T}_e)} + h_e^{\frac{1}{2}} |\delta \boldsymbol{\eta}^{j+1}|_{H^2(\mathcal{T}_e)} \\ &= h_e^{-\frac{1}{2}} |\delta \boldsymbol{\eta}^{j+1}|_{H^1(\mathcal{T}_e)} + h_e^{\frac{1}{2}} |\delta \mathbf{u}^{j+1}|_{H^2(\mathcal{T}_e)}, \end{aligned} \quad (40)$$

since $\mathbf{u}_I^j$ is a linear polynomial. An application of Taylor's theorem yields

$$\delta \mathbf{u}^{j+1} = \frac{1}{k_{j+1}} \left(k_{j+1} \dot{\mathbf{u}}^j + \int_{t_j}^{t_{j+1}} (t_{j+1} - t) \ddot{\mathbf{u}}(t) \, dt \right).$$

Thus,

$$|\delta \mathbf{u}^{j+1}|_{\mathbf{H}^2(\mathcal{T}_e)} \leq |\dot{\mathbf{u}}^j|_{\mathbf{H}^2(\mathcal{T}_e)} + \int_{t_j}^{t_{j+1}} |\ddot{\mathbf{u}}(t)|_{\mathbf{H}^2(\mathcal{T}_e)} dt. \quad (41)$$

Also, using relation derived in Eq. (37), we find

$$|\delta \boldsymbol{\eta}^{j+1}|_{\mathbf{H}^1(\mathcal{T}_e)} = \frac{1}{k_{j+1}} |\boldsymbol{\eta}^{j+1} - \boldsymbol{\eta}^j|_{\mathbf{H}^1(\mathcal{T}_e)} \lesssim h_e |\dot{\mathbf{u}}^j|_{\mathbf{H}^2(\mathcal{T}_e)} + \int_{t_j}^{t_{j+1}} |\ddot{\mathbf{u}}(t)|_{\mathbf{H}^1(\mathcal{T}_e)} dt. \quad (42)$$

Combining the relations obtained in Eqs. (40), (41), and (42), we obtain

$$|\delta \boldsymbol{\eta}^{j+1}|_{\mathbf{H}^1(e)} \lesssim h_e^{\frac{1}{2}} \|\dot{\mathbf{u}}^j\|_{\mathbf{H}^2(\mathcal{T}_e)} + h_e^{-\frac{1}{2}} \int_{t_j}^{t_{j+1}} |\ddot{\mathbf{u}}(t)|_{\mathbf{H}^1(\mathcal{T}_e)} dt + h_e^{\frac{1}{2}} \int_{t_j}^{t_{j+1}} |\ddot{\mathbf{u}}(t)|_{\mathbf{H}^2(\mathcal{T}_e)} dt. \quad (43)$$

Thus, Eq. (39) stems down to

$$\begin{aligned} & \sum_{j=1}^{N-1} \sum_{e \in \mathcal{E}_h^0} \|\mathbf{u}^j - \mathbf{u}_h^j\|_{\mathbf{L}^2(e)} \|\boldsymbol{\sigma}(\mathbf{u}^j - \mathbf{u}_I^j) - \boldsymbol{\sigma}(\mathbf{u}^{j+1} - \mathbf{u}_I^{j+1})\|_{\mathbf{L}^2(e)} \\ & \lesssim \sum_{j=1}^{N-1} \left\| \mathbf{e}^j \left\| \left(h^2 k_{j+1}^2 \|\dot{\mathbf{u}}^j\|_{\mathbf{H}^2(\Omega)}^2 + k_{j+1}^2 \left(\int_{t_j}^{t_{j+1}} |\ddot{\mathbf{u}}(t)|_{\mathbf{H}^1(\Omega)} dt \right)^2 + h^2 k_{j+1}^2 \left(\int_{t_j}^{t_{j+1}} |\ddot{\mathbf{u}}(t)|_{\mathbf{H}^2(\Omega)} dt \right)^2 \right) \right\| \right\| \\ & \lesssim \sum_{j=1}^{N-1} \left\| \mathbf{e}^j \left\| \left(h k_{j+1} \|\dot{\mathbf{u}}^j\|_{\mathbf{H}^2(\Omega)} + k_{j+1} \int_{t_j}^{t_{j+1}} |\ddot{\mathbf{u}}(t)|_{\mathbf{H}^1(\Omega)} dt + h k_{j+1} \int_{t_j}^{t_{j+1}} |\ddot{\mathbf{u}}(t)|_{\mathbf{H}^2(\Omega)} dt \right) \right\| \right\|. \end{aligned}$$

Now, we will bound the second term in the relation (38) in the following way:

$$\begin{aligned} & \sum_{j=1}^N \sum_{e \in \mathcal{E}_h^C} \int_e k_j (\sigma_n(\mathbf{u}^j - \mathbf{u}_I^j)) (\delta \mathbf{u}_I^j - \delta \mathbf{u}_h^j)_n d\sigma \\ & = \sum_{j=1}^N \sum_{e \in \mathcal{E}_h^C} \int_e (\sigma_n(\mathbf{u}^j - \mathbf{u}_I^j)) ((\mathbf{u}_I^j - \mathbf{u}_h^j) - (\mathbf{u}_I^{j-1} - \mathbf{u}_h^{j-1}))_n d\sigma. \end{aligned}$$

Next, we reformulate the term in the last relation by

$$\begin{aligned} & \sum_{j=1}^N \sum_{e \in \mathcal{E}_h^C} \int_e k_j (\sigma_n(\mathbf{u}^j - \mathbf{u}_I^j)) (\delta \mathbf{u}_I^j - \delta \mathbf{u}_h^j)_n d\sigma \\ & = \sum_{e \in \mathcal{E}_h^C} \int_e \sigma_n(\mathbf{u}^N - \mathbf{u}_I^N) (\mathbf{u}_I^N - \mathbf{u}_h^N)_n d\sigma + \sum_{e \in \mathcal{E}_h^C} \int_e \sigma_n(\mathbf{u}^1 - \mathbf{u}_I^1) (\mathbf{u}_I^0 - \mathbf{u}_h^0)_n d\sigma \\ & + \sum_{j=1}^{N-1} \sum_{e \in \mathcal{E}_h^C} \int_e (\sigma_n(\mathbf{u}^j - \mathbf{u}_I^j) - \sigma_n(\mathbf{u}^{j+1} - \mathbf{u}_I^{j+1})) (\mathbf{u}_I^j - \mathbf{u}_h^j)_n d\sigma. \end{aligned}$$

Since $\mathbf{u}_I^0 = \mathbf{u}_h^0$, we obtain

$$\sum_{j=1}^N \sum_{e \in \mathcal{E}_h^C} \int_e k_j (\sigma_n(\mathbf{u}^j - \mathbf{u}_I^j)) (\delta \mathbf{u}_I^j - \delta \mathbf{u}_h^j)_n d\sigma$$

$$\begin{aligned}
 &= \sum_{e \in \mathcal{E}_h^C} \int_e \sigma_n(\mathbf{u}^N - \mathbf{u}_I^N)(\mathbf{u}_I^N - \mathbf{u}_h^N)_n \, d\sigma \\
 &+ \sum_{j=1}^{N-1} \sum_{e \in \mathcal{E}_h^C} \int_e (\sigma_n(\mathbf{u}^j - \mathbf{u}_I^j) - \sigma_n(\mathbf{u}^{j+1} - \mathbf{u}_I^{j+1}))(\mathbf{u}_I^j - \mathbf{u}_h^j)_n \, d\sigma \\
 &= Q_1 + Q_2,
 \end{aligned}$$

where

$$\begin{aligned}
 Q_1 &= \sum_{e \in \mathcal{E}_h^C} \int_e \sigma_n(\mathbf{u}^N - \mathbf{u}_I^N)(\mathbf{u}_I^N - \mathbf{u}_h^N)_n \, d\sigma, \\
 Q_2 &= \sum_{j=1}^{N-1} \sum_{e \in \mathcal{E}_h^C} \int_e (\sigma_n(\mathbf{u}^j - \mathbf{u}_I^j) - \sigma_n(\mathbf{u}^{j+1} - \mathbf{u}_I^{j+1}))(\mathbf{u}_I^j - \mathbf{u}_h^j)_n \, d\sigma.
 \end{aligned}$$

For the term Q_1 , applying the Cauchy-Schwarz inequality, trace inequality, interpolation estimate, Poincaré-Wirtinger inequality, and inverse inequality, we find

$$\begin{aligned}
 Q_1 &= \sum_{e \in \mathcal{E}_h^C} \int_e \sigma_n(\mathbf{u}^N - \mathbf{u}_I^N)(\mathbf{u}_I^N - \mathbf{u}_h^N)_n \, d\sigma \\
 &\leq \sum_{e \in \mathcal{E}_h^C} |\mathbf{u}^N - \mathbf{u}_I^N|_{H^1(e)} \|(\mathbf{u}_I^N - \mathbf{u}_h^N)_n\|_{L^2(e)} \\
 &\lesssim \sum_{e \in \mathcal{E}_h^C} h_e^{\frac{1}{2}} |\mathbf{u}^N|_{H^2(\mathcal{T}_e)} h_e |(\mathbf{u}_I^N - \mathbf{u}_h^N)_n|_{H^1(e)} \\
 &\lesssim \sum_{e \in \mathcal{E}_h^C} h_e |\mathbf{u}^N|_{H^2(\mathcal{T}_e)} |\mathbf{u}_I^N - \mathbf{u}_h^N|_{H^1(\mathcal{T}_e)} \\
 &\lesssim h |\mathbf{u}^N|_{H^2(\Omega)} \|\boldsymbol{\theta}^N\|.
 \end{aligned}$$

For the term Q_2 , using the Cauchy-Schwarz inequality, Poincaré-Wirtinger inequality, and inverse inequality, we get

$$\begin{aligned}
 Q_2 &= \sum_{j=1}^{N-1} \sum_{e \in \mathcal{E}_h^C} \int_e (\sigma_n(\mathbf{u}^j - \mathbf{u}_I^j) - \sigma_n(\mathbf{u}^{j+1} - \mathbf{u}_I^{j+1}))(\mathbf{u}_I^j - \mathbf{u}_h^j)_n \, d\sigma \\
 &\leq \sum_{j=1}^{N-1} \sum_{e \in \mathcal{E}_h^C} \left(\|\sigma_n(\mathbf{u}^j - \mathbf{u}_I^j) - \sigma_n(\mathbf{u}^{j+1} - \mathbf{u}_I^{j+1})\|_{L^2(e)} \|(\mathbf{u}_I^j - \mathbf{u}_h^j)_n\|_{L^2(e)} \right) \\
 &\lesssim \sum_{j=1}^{N-1} \sum_{e \in \mathcal{E}_h^C} \left(h_e \|\sigma_n(\mathbf{u}^j - \mathbf{u}_I^j) - \sigma_n(\mathbf{u}^{j+1} - \mathbf{u}_I^{j+1})\|_{L^2(e)} |(\mathbf{u}_I^j - \mathbf{u}_h^j)_n|_{H^1(e)} \right) \\
 &\lesssim \sum_{j=1}^{N-1} \sum_{e \in \mathcal{E}_h^C} \left(h_e^{\frac{1}{2}} \|\sigma_n(\mathbf{u}^j - \mathbf{u}_I^j) - \sigma_n(\mathbf{u}^{j+1} - \mathbf{u}_I^{j+1})\|_{L^2(e)} |\mathbf{u}_I^j - \mathbf{u}_h^j|_{H^1(\mathcal{T}_e)} \right)
 \end{aligned}$$

$$\begin{aligned}
&\lesssim \sum_{j=1}^{N-1} \left(\left(\sum_{e \in \mathcal{E}_h^C} h_e \|\sigma_n(\mathbf{u}^j - \mathbf{u}_I^j) - \sigma_n(\mathbf{u}^{j+1} - \mathbf{u}_I^{j+1})\|_{L^2(e)}^2 \right)^{\frac{1}{2}} \|\boldsymbol{\theta}^j\| \right) \\
&\lesssim \sum_{j=1}^{N-1} \left(\left(\sum_{e \in \mathcal{E}_h^C} h_e |(\mathbf{u}^j - \mathbf{u}_I^j) - (\mathbf{u}^{j+1} - \mathbf{u}_I^{j+1})|_{\mathbf{H}^1(e)}^2 \right)^{\frac{1}{2}} \|\boldsymbol{\theta}^j\| \right) \\
&= \sum_{j=1}^{N-1} \left(\|\boldsymbol{\theta}^j\| \left(\sum_{e \in \mathcal{E}_h^C} h_e k_{j+1}^2 |\delta \boldsymbol{\eta}^{j+1}|_{\mathbf{H}^1(e)}^2 \right)^{\frac{1}{2}} \right).
\end{aligned}$$

Utilizing the bound of $|\delta \boldsymbol{\eta}^{j+1}|_{\mathbf{H}^1(e)}$ derived in Eq. (43), we obtain

$$Q_2 \lesssim \sum_{j=1}^{N-1} \|\boldsymbol{\theta}^j\| \left(h k_{j+1} \|\dot{\mathbf{u}}^j\|_{\mathbf{H}^2(\Omega)} + k_{j+1} \int_{t_j}^{t_{j+1}} |\ddot{\mathbf{u}}(t)|_{\mathbf{H}^1(\Omega)} dt + h k_{j+1} \int_{t_j}^{t_{j+1}} |\ddot{\mathbf{u}}(t)|_{\mathbf{H}^2(\Omega)} dt \right). \quad (44)$$

Observe that for $j = 1, 2, \dots, N$, we have

$$\begin{aligned}
\|\boldsymbol{\theta}^j\| &\leq \|\boldsymbol{\eta}^j\| + \|\mathbf{e}^j\| \\
&\lesssim h |\mathbf{u}^j|_{\mathbf{H}^2(\Omega)} + \|\mathbf{e}^j\|.
\end{aligned} \quad (45)$$

Inserting Eq. (45) in relation (44), we obtain

$$\begin{aligned}
Q_2 &\lesssim \sum_{j=1}^{N-1} \|\mathbf{e}^j\| \left(h k_{j+1} \|\dot{\mathbf{u}}^j\|_{\mathbf{H}^2(\Omega)} + k_{j+1} \int_{t_j}^{t_{j+1}} |\ddot{\mathbf{u}}(t)|_{\mathbf{H}^1(\Omega)} dt + h k_{j+1} \int_{t_j}^{t_{j+1}} |\ddot{\mathbf{u}}(t)|_{\mathbf{H}^2(\Omega)} dt \right) \\
&\quad + \sum_{j=1}^{N-1} \left(h^2 k_{j+1} \|\dot{\mathbf{u}}^j\|_{\mathbf{H}^2(\Omega)} |\mathbf{u}^j|_{\mathbf{H}^2(\Omega)} + h k_{j+1} \int_{t_j}^{t_{j+1}} |\ddot{\mathbf{u}}(t)|_{\mathbf{H}^1(\Omega)} dt |\mathbf{u}^j|_{\mathbf{H}^2(\Omega)} \right. \\
&\quad \left. + h^2 k_{j+1} \int_{t_j}^{t_{j+1}} |\ddot{\mathbf{u}}(t)|_{\mathbf{H}^2(\Omega)} dt |\mathbf{u}^j|_{\mathbf{H}^2(\Omega)} \right).
\end{aligned}$$

Thus, the relation (38) reduces to

$$\begin{aligned}
\|\mathbf{e}^N\|^2 &\lesssim h \max_n |\dot{\mathbf{u}}^n|_{\mathbf{H}^2(\Omega)} \max_n \|\mathbf{e}^n\| \sum_{j=1}^N k_j + h k \max_n \|\mathbf{e}^n\| \left(\int_{\mathcal{I}} |\ddot{\mathbf{u}}(t)|_{\mathbf{H}^1(\Omega)} dt \right) \\
&\quad + k \max_n \|\mathbf{e}^n\| \int_{\mathcal{I}} |\ddot{\mathbf{u}}(t)|_{\mathbf{H}^2(\Omega)} dt + h^2 \max_n |\dot{\mathbf{u}}^n|_{\mathbf{H}^2(\Omega)} \max_n |\mathbf{u}^n|_{\mathbf{H}^2(\Omega)} \sum_{j=1}^N k_j \\
&\quad + h k \max_n |\mathbf{u}^n|_{\mathbf{H}^2(\Omega)} \int_{\mathcal{I}} |\ddot{\mathbf{u}}(t)|_{\mathbf{H}^2(\Omega)} dt + h^2 k |\mathbf{u}^n|_{\mathbf{H}^2(\Omega)} \int_{\mathcal{I}} |\ddot{\mathbf{u}}(t)|_{\mathbf{H}^2(\Omega)} dt \\
&\quad + \|g_a\|_{L^\infty(\Gamma_C)} h^2 \sum_{j=1}^N k_j \max_n |\dot{\mathbf{u}}^n|_{\mathbf{H}^2(\Omega)} + h \max_n \|\mathbf{e}^n\| \max_n |\dot{\mathbf{u}}^{n-1}|_{\mathbf{H}^2(\Omega)} \\
&\quad + \max_n \|\mathbf{e}^n\| \sum_{j=1}^N \left(k_j \int_{t_{j-1}}^{t_j} |\ddot{\mathbf{u}}(t)| dt \right) + h^2 \max_n \|\mathbf{u}^n\|_{\mathbf{H}^2(\Omega)}^2 + h^2 \max_n \|\dot{\mathbf{u}}^n\|_{\mathbf{H}^2(\Omega)}^2.
\end{aligned} \quad (46)$$

Using the identity $ab \leq \frac{a^2}{2} + \frac{b^2}{2}$, we obtain

$$\begin{aligned} \|e^N\|^2 &\lesssim h \max_n |\dot{u}^n|_{H^2(\Omega)} \max_n \|e^n\| \sum_{j=1}^N k_j + h k \max_n \|e^n\| \left(\int_{\mathcal{I}} |\ddot{u}(t)|_{H^1(\Omega)} dt \right) \\ &\quad + k \max_n \|e^n\| \int_{\mathcal{I}} |\ddot{u}(t)|_{H^2(\Omega)} dt + h^2 \max_n |\dot{u}^n|_{H^2(\Omega)}^2 + h^2 \max_n |u^n|_{H^2(\Omega)}^2 \\ &\quad + h^2 \max_n |u^n|_{H^2(\Omega)}^2 + k^2 \left(\int_{\mathcal{I}} |\ddot{u}(t)|_{H^2(\Omega)} dt \right)^2 + h^4 |u^n|_{H^2(\Omega)}^2 \\ &\quad + k^2 \left(\int_{\mathcal{I}} |\ddot{u}(t)|_{H^2(\Omega)} dt \right)^2 + \|g_a\|_{L^\infty(\Gamma_C)} h^2 \sum_{j=1}^N k_j \max_n |\dot{u}^n|_{H^2(\Omega)} \\ &\quad + h \max_n \|e^n\| \max_n |\dot{u}^{n-1}|_{H^2(\Omega)}. \end{aligned} \quad (47)$$

Let $M = \max_n \|e^n\|$. Then, we have

$$\begin{aligned} M^2 &\lesssim M(h \|u^n\|_{L^\infty(\mathcal{I}; H^2(\Omega))} + h \|\dot{u}^n\|_{L^\infty(\mathcal{I}; H^2(\Omega))} + h^2 \|\ddot{u}\|_{L^1(\mathcal{I}; V)} + k^2 \|\ddot{u}\|_{L^1(\mathcal{I}; V)} \\ &\quad + k \|\ddot{u}\|_{L^1(\mathcal{I}; H^2(\Omega))}) + h^2 \|u\|_{L^\infty(\mathcal{I}; H^2(\Omega))}^2 + k^2 \|\ddot{u}\|_{L^1(\mathcal{I}; V)}^2 + h^2 \|\dot{u}\|_{L^\infty(\mathcal{I}; H^2(\Omega))}^2 \\ &\quad + k^2 \|\ddot{u}\|_{L^1(\mathcal{I}; H^2(\Omega))}^2. \end{aligned}$$

Using the identity if $x^2 \leq ax + b$ where $x, a, b > 0$ then $x \leq a + b^{\frac{1}{2}}$, we obtain

$$\max_n \|e^n\| \lesssim h + k.$$

This completes the proof.

5 Numerical Experiments

In this section, we present the numerical experiment to examine the convergence characteristics of the fully discrete scheme on uniform triangular grids. The experiment has been carried out in MATLAB_R2020B. In order to transform the variational inequality (27) into an equality, we introduce the Lagrange multiplier $\lambda_h \in L^\infty(\Gamma_C)$ using the Uzawa algorithm [16]. With this, the variational inequality (27) is equivalent to

$$a_h(u_h^n, v_h) + \int_{\Gamma_C} g_a \lambda_{h\tau} \cdot v_{h\tau} d\sigma = (I^n, v_h), \quad \forall v_h \in \mathcal{K}_{CR}^h.$$

The following is the concise overview of the Uzawa algorithm implemented at each time step t_n .

Step 1: Initialize $\lambda_h^0 = \mathbf{0}$ and compute the solution $[u_h^n]^0 \in \mathcal{K}_{CR}^h$ of the discrete variational equality:

$$a_h([u_h^n]^0, v_h) = (I^n, v_h), \quad \forall v_h \in \mathcal{K}_{CR}^h.$$

Step 2: Define a projection map P as

$$P(\chi) := \sup(-1, \inf(1, \chi)), \quad \forall \chi \in L^\infty(\Gamma_C).$$

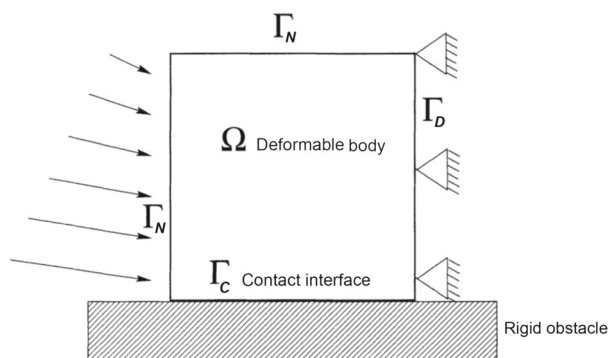


Fig. 1 Physical setting for Example 1

With the help of this projection operator, update the Lagrange multiplier $\lambda_h^k, k = 1, 2, \dots$, as

$$\lambda_h^k = P(\lambda_h^{k-1} + \tilde{\rho}[u_h^n]^{k-1})$$

and solve for $[u_h^n]^k$ from the following equation:

$$a_h([u_h^n]^k, v_h) = (l^n, v_h) - \int_{\Gamma_C} \lambda_{h\tau}^k \cdot v_{h\tau} \, ds, \quad \forall v_h \in \mathcal{K}_{CR}^h,$$

where $\tilde{\rho}$ is a positive constant and is chosen to be 1 in the following example.

Step 3: The iteration process is terminated when $|[u_h^n]^k - [u_h^n]^{k-1}| < \epsilon$ where ϵ denotes the error tolerance.

Now, we present the numerical outcome of a two-dimensional example. Therein, we employ a fully discrete scheme and approximate the derivatives by backward Euler difference. The domain under consideration is a square domain and the triangles are refined uniformly at each iteration. We set the error tolerance ϵ as 10^{-8} and the penalty parameter ρ is chosen as 10.

Example 1 In this model problem motivated from [17], we consider the deformation of the domain $\Omega := (0, 4) \times (0, 4)$ which represents the cross-section of a three-dimensional linearly elastic body subjected to certain traction forces. The body is clamped on $\Gamma_D := \{4\} \times (0, 4)$, and therefore, the displacement field vanishes there. Surface traction forces act on the part of the Neumann boundary $\{0\} \times (0, 4)$ whereas, the part $(0, 4) \times \{4\}$ is traction free. Thus, $\Gamma_N := (\{0\} \times (0, 4)) \cup ((0, 4) \times \{4\})$. The body is in contact with the rigid foundation on the part of the contact boundary $\Gamma_C := (0, 4) \times \{0\}$. We consider discretizing the space V using linear elements and dividing the time interval I into uniform partitions. For our calculations, we have employed the following data, where the unit daN/mm^2 corresponds to deca-newtons per square millimeter:

$$\begin{aligned} E &= 200, \quad \nu = 0.3, \quad f = (0, 0) \text{ daN/mm}^2, \\ g &= (0.02(5 - y)t, -0.01t) \text{ daN/mm}^2, \\ g_a &= 0.0012 \text{ daN/mm}^2, \quad u^0 = \mathbf{0} \text{ m}, \quad T = 1 \text{ s}. \end{aligned}$$

Table 1 Error and order of convergence with respect to the mesh parameter h for Example 1

N	DOF	h	Total error	Order
40	28	2^{-1}	2.512×10^{-4}	–
80	104	2^{-2}	1.431×10^{-4}	0.8579
160	400	2^{-3}	7.460×10^{-5}	0.9657
320	1568	2^{-4}	4.030×10^{-5}	0.9006
640	6208	2^{-5}	2.2210×10^{-5}	0.8939

Table 2 Error and order of convergence with respect to the time parameter k for Example 1

DOF	N	k	Total error	Order
104	20	0.05	1.411×10^{-4}	–
400	40	0.025	7.410×10^{-4}	0.9292
1568	80	0.0125	4.010×10^{-5}	0.8859
6208	160	0.00625	2.1220×10^{-5}	0.9182
12416	320	0.003125	1.0810×10^{-5}	0.9731

The Lamé parameters μ and λ are computed by

$$\mu = \frac{E}{2(1+\nu)} \quad \text{and} \quad \lambda = \frac{E\nu}{1-\nu^2}.$$

As the exact solution $\mathbf{u}$ is unknown in the above example, we calculate the energy norm error by calculating the difference between the discrete solutions obtained on the consecutive meshes. The pictorial representation of the above example is depicted in Fig. 1. Tables 1 and 2 depict the energy norm error and its order of convergence for the spatial variable h and time variable k . In particular, the convergence order for the spatial variable h and temporal variable k is shown for a fixed ratio $\frac{k}{h} = 0.1$. Here, **DOF** represents the degrees of freedom, and N indicates the number of time steps. We observe that the numerical convergence rates are near 1, which is in accordance with the theoretical results derived in Sect. 4.

Declarations

Conflict of Interest The authors declare no potential conflict of interest.

References

1. Alart, P., Curnier, A.: A mixed formulation for frictional contact problems prone to Newton like solution methods. *Comput. Methods Appl. Mech. Eng.* **92**(3), 353–375 (1991)
2. Andersson, L.E.: A quasistatic frictional problem with normal compliance. *Nonlinear Anal. Theo. Method. Appl.* **16**(4), 347–369 (1991)
3. Andersson, L.E.: Existence results for quasistatic contact problems with Coulomb friction. *Appl. Math. Opt.* **42**, 169–202 (2000)
4. Andersson, L.E., Klarbring, A.: A review of the theory of static and quasi-static frictional contact problems in elasticity. *Philos. Trans. R. Soc. London. Ser. A: Math. Phys. Eng. Sci.* **359**(1789), 2519–2539 (2001)
5. Barboteu, M., Han, W., Sofonea, M.: Numerical analysis of a bilateral frictional contact problem for linearly elastic materials. *IMA J. Numer. Anal.* **22**(3), 407–436 (2002)
6. Capatina, A.: Existence and uniqueness results. *Var. Inequal. Frict. Contact Probl.* **3**, 31–82 (2014)

7. Carstensen, C., Köhler, K.: Nonconforming FEM for the obstacle problem. *IMA J. Numer. Anal.* **37**(1), 64–93 (2017)
8. Ciarlet, P.G.: *The Finite Element Method for Elliptic Problems*. SIAM, Philadelphia (2002)
9. Cocou, M.: A variational inequality and applications to quasistatic problems with Coulomb friction. *Appl. Anal.* **97**(8), 1357–1371 (2018)
10. Cocu, M.: Existence of solutions of Signorini problems with friction. *Int. J. Eng. Sci.* **22**(5), 567–575 (1984)
11. Cocu, M., Pratt, E., Raous, M.: Formulation and approximation of quasistatic frictional contact. *Int. J. Eng. Sci.* **34**(7), 783–798 (1996)
12. Djoko, J.K., Ebobisse, F., McBride, A.T., Reddy, B.: A discontinuous Galerkin formulation for classical and gradient plasticity—Part 1: formulation and analysis. *Comput. Methods Appl. Mech. Eng.* **196**(37/38/39/40), 3881–3897 (2007)
13. Djoko, J.K., Ebobisse, F., McBride, A.T., Reddy, B.: A discontinuous Galerkin formulation for classical and gradient plasticity. Part 2: algorithms and numerical analysis. *Comput. Methods Appl. Mech. Eng.* **197**(1/2/3/4), 1–21 (2007)
14. Duvaut, G., Lions, J.L.: *Inequalities in Mechanics and Physics*. Springer Science & Business Media, Berlin (2008)
15. Fernández, J.R., Hild, P.: A priori and a posteriori error analyses in the study of viscoelastic problems. *J. Comput. Appl. Math.* **225**(2), 569–580 (2009)
16. Glowinski, R.: *Lectures on Numerical Methods for Non-linear Variational Problems*. Springer Science & Business Media, Berlin (2008)
17. Han, W., Sofonea, M.: *Quasistatic Contact Problems in Viscoelasticity and Viscoplasticity*. American Mathematical Society, Providence, R.I. (2002)
18. Hansbo, P., Larson, M.G.: Discontinuous Galerkin and the Crouzeix-Raviart element: application to elasticity. *ESAIM: Math. Modell. Numer. Anal.* **37**(1), 63–72 (2003)
19. Kesavan, S.: Topics in functional analysis and applications. *Math. Comp. Sim.* **32**(4), 424 (1990)
20. Kikuchi, N., Oden, J.T.: *Contact Problems in Elasticity: a Study of Variational Inequalities and Finite Element Methods*. SIAM, Philadelphia (1988)
21. Klarbring, A., Mikelić, A., Shillor, M.: Frictional contact problems with normal compliance. *Int. J. Eng. Sci.* **26**(8), 811–832 (1988)
22. Klarbring, A., Mikelić, A., Shillor, M.: On friction problems with normal compliance. *Nonlinear Anal.* **13**(8), 935–955 (1989)
23. Rocca, R.: Existence of a solution for a quasistatic problem of unilateral contact with local friction. *Comptes Rendus de l'Académie des Sciences-Series I-Mathematics* **328**(12), 1253–1258 (1999)
24. Rocca, R., Cocu, M.: Existence and approximation of a solution to quasistatic Signorini problem with local friction. *Int. J. Eng. Sci.* **39**(11), 1233–1255 (2001)
25. Scott, L.R., Zhang, S.: Finite element interpolation of nonsmooth functions satisfying boundary conditions. *Math. Comput.* **54**(190), 483–493 (1990)
26. Shillor, M., Sofonea, M., Telega, J.J.: *Models and Analysis of Quasistatic Contact: Variational Methods*, vol. 655. Springer Science & Business Media, Berlin (2004)
27. Sofonea, M., Matei, A.: Variational inequalities with applications: a study of antiplane frictional contact problems, In: *Advances in Mechanics and Mathematics*, vol. 18. Springer Science & Business Media, Berlin (2009)
28. Wang, F., Han, W., Cheng, X.: Discontinuous Galerkin methods for solving a quasistatic contact problem. *Numer. Math.* **126**(4), 771–800 (2014)

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.